

Binding vs Co-reference: On Logophoricity in German*

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Abstract The present study investigates whether German has logophoric anaphors. Drawing on new experimental evidence, I show that non-local anaphors are accepted by some speakers of German if they i) are c-commanded by their antecedent and ii) occur in a logophoric context. These anaphors hence show properties associated with binding as well as with logophoric co-reference. To solve this paradox, I propose that anaphors in German always require syntactic binding, but can be assigned an interpretation either by means of semantic binding or by means of logophoric co-reference.

Keywords anaphors, logophoricity, binding, German

1 Introduction

Pronouns are nominals with quite uninformative lexical entries. How they nevertheless take on a specific meaning when used in discourse has long been one of the major issues in generative linguistics. In this paper, I will be concerned with the process of anaphora or co-indexation, that is, with the mechanisms by which the interpretation of a pronoun, meaning a pronominal or an anaphor, can be made dependent on the interpretation of another NP, its antecedent. There is broad consensus that we can distinguish between two types of anaphora:

- (1a) After Maryam_i had checked the manual, she_i turned on the engine.¹
- (1b) [Every man]_i thinks that he_i is a good father.

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1 I use co-indexation as a notational device both for binding and for co-reference, with no theoretical implications. Glossing conventions used in this paper: GEN = genitive, DAT = dative, ACC = accusative, SUBJ = subjunctive, PRE = verbal prefix separated from the stem. All grammaticality judgments in cited examples are in the original.

In (1a), *she* simply picks up the same referent as *Maryam*. We can hence say that *she* co-refers with *Maryam*. The following reformulation is syntactically awkward, but truth-conditionally correct:

- (2) After Maryam_i had checked the manual, Maryam_i turned on the engine.

On the other hand, the same does not hold for (1b):

- (3) Every man thinks that every man is a good father.

Sentence (3), while fully grammatical, does not capture the original meaning of (1b). Rather, we can reformulate the latter roughly as follows:

- (4) For every x such that x is a man, x thinks that x is a good father.

In (1b), *he* hence does not refer to the set of all men, but is bound to a variable x, such that whenever x varies, so does the interpretation of *he*.

Co-reference and binding are two fundamental mechanisms governing the assignment of meaning to pronouns, potentially even located in distinct areas of the grammar (Grodzinsky and Reinhart 1993, Reuland 2001). One of the most important insights of binding theory has been to link binding, but not co-reference to specific syntactic conditions: Semantic binding requires syntactic binding, that is, in order to be semantically bound as described, a pronoun must be c-commanded by the co-indexed DP (e.g. Heim and Kratzer 1998).²

As demonstrated above, pronominals can have both a bound and a co-referential use, while in early binding theory anaphors were claimed to allow for a bound use only (Chomsky 1981). However, there is now considerable consensus that anaphors can in fact be used co-referentially if they appear in a particular pragmatic, namely logophoric context. While this notion will be introduced in more detail in chapter 2, we can give a first approximate definition by saying that a logophoric anaphor must co-refer with the individual from whose perspective the sentence containing the anaphor is presented. Logophoricity hence enables anaphors to cross the line between binding and co-reference, such that they show properties associated with the one or the other mechanism depending on the context in which they occur.

The present study takes as its starting point the question whether such logophoric anaphors can be observed in one particular language, namely German. While it is widely assumed that anaphors

2 As pointed out by Heim and Kratzer (1998:261), syntactic binders need to be in an A-position. However, since c-commanding antecedents in A'-positions will not be discussed in the present study, I will ignore this aspect in the following.

in German always require local binding and do not allow for a logophoric use, I will provide new experimental evidence challenging this view. Gathering a reliable understanding of the binding behaviour of German anaphors is highly relevant in and of itself, especially given that binding is frequently used as a syntactic diagnostic for phenomena such as constituency, movement and reconstruction. Moreover, on a theoretical level, I will show that the German data force us to call into question the clear-cut distinction between co-reference and binding introduced above. In particular, I will argue that decoupling syntactic and semantic binding helps us understand the behaviour of anaphors in German, and potentially in other languages as well.

I will now define more precisely the scope of this study. German has only one morphologically distinct reflexive pronoun, namely *sich*, which I will be exclusively concerned with. *Sich* is a third-person pronoun not specified for gender and number and ambiguous between dative and accusative. Insofar as in the plural, *sich* is in some contexts ambiguous between a reflexive and a reciprocal use, only the reflexive interpretation will be considered:

- (5) Maryam_i und Lisa_j mögen sich_{i+j}.
 Maryam_i and Lisa_j like themselves_{i+j}/each other_{i+j}.

The following will hence be excluded from the analysis. First, I do not look at the reciprocal *einander*. Secondly, I ignore the possessive *ihr/sein eigener*, which might have a logophoric use as in English (*her/his own*), French (*sa/son propre*), Italian (*propria/o*), and others. Thirdly, German reflexives also occur in the complex form *sich selbst*. *Selbst* can be analysed as an intensifier that, among other functions,³ adds emphasis to a reflexive occurring in the presence of a contextually salient alternative (Eckardt 2001, Gast 2006, Gast and Siemund 2006):

- (6a) Lisa ist wütend auf sich.
 Lisa is angry at herself.
- (6b) Lisa ist nicht wütend auf ihre Tochter, sondern auf sich selbst.
 Lisa is not angry at her daughter but at herself self.

3 I will gloss over the fact that *selbst* can also serve as a reflexivizer when adjoining to pronominal forms for which a reflexive equivalent is not available, particularly in adnominal genitives (Zifonun 2003):

- (i) Sie betrachtete ein Bild ihrer selbst.
 She looked.at a picture her.GEN self.
 She looked at a picture of herself.

Lisa is not angry at her daughter, but at herself.

Since *selbst* might affect the binding behaviour of anaphors, I will limit myself to the simple form *sich*.

Finally, throughout this paper, I will steer away from theoretical debates whenever these seem irrelevant for the question under consideration. I will start out with a view on anaphor binding that is based on the notion of c-command and a rudimentary notion of locality, following canonical binding theory as proposed in the GB framework. Since the 1990s, a competing approach has emerged which regards binding as a property not holding on anaphors and their antecedents, but on predicates (Reinhart and Reuland 1991, 1993, Pollard and Sag 1992). Such theories depart from traditional accounts in two major ways: First, predicate-based approaches propose to restrict the scope of binding theory to anaphors in co-argumental position, as will be outlined in chapter 2. This idea, while not uncontroversial, has proven fruitful, and I will draw on it repeatedly. Secondly, Reinhart and Reuland's (1991, 1993) theory rests crucially on the distinction between morphologically simple SE-anaphors and morphologically complex SELF-anaphors. However, as Reuland (2006) concedes, German *sich* and *sich selbst* only superficially follow this pattern,⁴ which makes the application of Reinhart and Reuland's theory in its pure form to German rather cumbersome. Hence, I will mostly rely on the more traditional, slimmer and simpler approach, implement insights from predicate-based theories when adequate and hope that the generalizations I arrive at will be amenable to any theoretical framework.

I will proceed as follows. After reviewing the literature on logophoric anaphors in chapter 2, I will then in chapter 3 present experimental data on logophoricity in German. While the results *prima facie* seem to support the hypothesis that German anaphors can have a logophoric use, in chapter 4 I will discuss several challenges to this interpretation of the findings, ultimately arriving at a more fine-grained proposal. Chapter 5 is a brief conclusion.

4 Reinhart and Reuland (1991, 1993) claim that SE-anaphors are obligatorily subject-oriented since they adjoin to I and receive their ϕ -features from the subject. However, this cannot be maintained for *sich*:

- (i) Hamid_i zeigt Lea_j ein Foto von sich_{i/j}.
Hamid_i shows Lea_j a photo of himself_i/herself_j.

See also Hendriks et al. (2014) for distributional differences between *sich* and the Dutch SE-anaphor *zich*.

2 Background

The following chapter reviews the literature on logophoric anaphors. 2.1 introduces the notion of logophoricity and demonstrates how a theory of logophoric anaphors can account for apparent violations of Binding Principle A, 2.2 outlines how to identify such logophoric anaphors and 2.3 summarises the literature on logophoric anaphors in German in particular.

2.1 From Principle A to Long-Distance Binding and Logophoricity

Canonical binding theory proposes that anaphors are subject to Principle A, which requires them to be bound in a local domain (Chomsky 1981). Under the definition of the relevant domain as the smallest category containing the anaphor, its governor and a subject, Principle A correctly predicts the following grammaticality effects:

- (7a) Aisha_i looks at herself_i in the mirror.
- (7b) *Aisha_i's parents liked the picture of herself_i.
- (7c) *He_i thought that Aisha liked himself_i.
- (7d) *I look at herself.

In (7a), the anaphor is locally bound, as required. In (7b), on the other hand, the antecedent does not c-command the anaphor, and in (7c), the antecedent is outside the local domain of the anaphor. In (7d), the anaphor lacks an antecedent altogether.

However, Principle A also makes several wrong predictions:

- (8a) Max_i boasted that the queen invited Lucie and him_i for a drink.
 - (8b) Max_i boasted that the queen invited Lucie and himself_i for a drink.
 - (8c) *Max_i boasted that the queen invited himself_i for a drink.
- (Reinhart and Reuland 1993:670)

- (9a) Engineers like you are rare.
- (9b) Engineers like yourself are rare.

In (8b), the antecedent is outside the local domain, while in (9b), the anaphor does not have an antecedent at all. Nevertheless, in both cases, the anaphoric form freely alternates with the pronominal. Additionally, the contrast between (8b) and (8c) remains unexplained.

A broad range of approaches has been developed to capture such apparent exemptions from Principle A (see for a recent overview Charnavel 2019:8–22). For the present purposes, it shall suffice to contrast two distinct strategies, glossing over some more fine-grained theoretical differences. On the one hand, it has been argued that seemingly exempt anaphors are in fact bound, violating Principle A only apparently. Such a position has been maintained by adjusting or parameterizing the binding domain (e.g. Chomsky 1981, Yang 1983, Koster 1984, Manzini and Wexler 1987), or by postulating covert movement of the anaphor to the antecedent’s domain, similar to clitic climbing (e.g. Lebeaux 1984, Pica 1987, Cole et al. 1990). For terminological clarity, I will refer to such theories as long-distance binding approaches, which maintain the idea that anaphors by their very nature require binding.

According to the alternative approach, anaphors are in fact exempt from binding if they are used logophorically⁵ (e.g. Kuno 1972, Cantrall 1974, Sells 1987, Zribi-Hertz 1989, Reinhart and Reuland 1991, Pollard and Sag 1992). Under this view, there are two types of anaphors: Plain anaphors need to be bound by a local antecedent, as predicted by (some version of) Principle A. Exempt or logophoric anaphors, on the other hand, rather behave like peculiar pronominals: Although superficially identical to plain anaphors, they are not bound, but instead refer back to one or several individuals from whose perspective the constituent⁶ they appear in is presented, regardless of their structural relation to this antecedent. Exempt anaphors are thus licensed only in logophoric contexts

5 The term logophor was originally coined to describe morphologically distinct pronominal markers in some African languages that refer back to the author of discourse in indirect speech (Hagège 1974) or more generally to someone whose thoughts or states of consciousness are presented (Clements 1975, Frajzyngier 1985, Ameka 2017), as demonstrated here for the logophor *yè* in Ewe:

- (i) Kofi be *yè*-dzo.
Kofi say LOG-leave.
Kofi_i said that he_{i/*j} left.
 - (ii) Kofi be *e*-dzo.
Kofi say 3SG-leave.
Kofi_i said that s/he_{*i/j} left.
- (Clements 1975:242)

Whether such specific logophors and logophorically used anaphors should be considered a unified phenomenon is matter of debate and has been called into question e.g. by Culy (1994, 1997). In the following, I will thus discuss logophoric anaphors only.

which are presented from the perspective of their antecedent and ungrammatical otherwise.

Borrowing Charnavel's (2019) terminology, I will refer to such perspective-holding antecedents as logophoric centres.

To exemplify this idea, we can look again at (8b), repeated here as (10):

(10) Max_i boasted that the queen invited Lucie and himself_i for a drink.

The antecedent *Max* is the logophoric centre and the embedded clause is the logophoric context, presented not from the speaker's, but from Max' perspective. In fact, the speaker might disagree:

(11) Max_i boasted that the queen invited Lucie and himself_i for a drink, but I know for a fact that he lied.

In consequence, the anaphor is exempt from binding and instead co-refers with *Max*.

As for more systematic evidence that exempt anaphors are licensed in logophoric contexts, this hypothesis entails that exempt anaphors should be unacceptable if their antecedent is inanimate, since qualifying as a logophoric centre requires having a mental perspective, which in turn requires animacy. Charnavel and Sportiche (2016) show that this prediction is borne out:

(12a) [The earth]_i is subject to the gravitational effect of the numerous satellites that revolve around it(*-self)_i.

(12b) Mary_i is subject to the influence of the many politicians that revolve around her(-self)_i.
(Charnavel and Sportiche 2016:41)

This fact remains unexplained under a purely syntactic long-distance approach and instead offers evidence that exempt anaphors are only licensed in contexts presented from the perspective of their antecedent.

Finally, in addition to these pragmatic constraints, exempt anaphors are also limited to certain syntactic positions:

(13a) Max_i boasted that the queen invited Lucie and himself_i for a drink.

(13b) * Max_i boasted that the queen invited himself_i for a drink.

6 Traditionally, logophoric contexts have been assumed to be propositional in nature and hence clausal, which has recently been challenged by Charnavel (2019). I will leave this question open.

As a first approximation, let us say that anaphors realized as objects cannot be exempt. This is demonstrated by the contrast in grammaticality in (13) – since in (13a), *himself* is only one conjunct making up the argument – as well as by the following example:

- (14a) He_i thought that Aisha liked the picture of himself_i.
 (14b) *He_i thought that Aisha liked himself_i.

To account for these effects, Reinhart and Reuland (1991, 1993; see also Pollard and Sag 1992 for a similar HPSG-based approach) propose to reformulate the Binding Principles A and B as follows: Nominals that are co-indexed with a co-argument must be realized as anaphors, and vice versa.⁷ This rules out the sentences in (15):

- (15a) *Aisha_i liked her_i.
 (15b) *Aisha_i liked herself_j.

Anaphors lacking a co-argument altogether do not fall in the scope of this rule and thus do not need to be bound. On the other hand, complements of verbal and adjectival predicates are predicted to be never exempt since they always find a co-argument in the subject.⁸

Although Reinhart and Reuland's theory covers a considerable range of data, it has also been met with criticism that I cannot address here (see e.g. Charnavel and Sportiche 2016, Charnavel 2019). What is relevant for our present purposes is that logophoric anaphors are restricted syntactically, with a rough, but fairly robust generalization being that they cannot be realized as objects. Whether the notion of co-argumenthood is helpful to derive more precise constraints on the syntactic distribution of exempt anaphors in German will be discussed in chapter 4.

7 I considerably simplify this theory, omitting in particular the role played by SE- and SELF-anaphors.

8 The subject position itself cannot host anaphors in languages such as German and English, regardless of exemptness, perhaps since it is construed with agreement (Rizzi 1990, Woolford 1999, but see Preminger 2020). Moreover, Reinhart and Reuland (1991, 1993) also regard adnominal complements as co-argumental if SpecDP is filled, predicting the following contrast:

- (i) Zadi_i liked the picture of herself_i.
 (ii) *Zadi_i liked Sue_j's picture of herself_i.

However, Keller and Asudeh (2001) could show experimentally that (ii) is in fact acceptable for speakers and propose that possessors are not true co-arguments of adnominal complements.

To sum up, there is broad consensus that in various languages⁹, anaphors can be exempt from Principle A if they are used logophorically. What remains controversial, among other things, is whether all instances of apparent exemption can be reduced to logophoricity, as argued recently by Charnavel (2019), or whether a subset of apparently exempt anaphors are in fact long-distance bound and hence unaffected by discourse constraints (e.g. Thráinsson 1991, Reinhart and Reuland 1991, 1993, Sigurjónsdóttir and Hyams 1992, Cole et al. 2006). I will return to this discussion in the conclusion of this study.

2.2 *Diagnostics for Logophoric Anaphors*

After having introduced the notion of logophoric anaphors, I will now outline how to identify them. I discuss three widely used diagnostics¹⁰ (e.g. Bouchard 1984, Lebeaux 1984, Charnavel and Sportiche 2016, Charnavel 2019) which are all based on one assumption: Logophoric anaphors are assigned an interpretation by means of co-reference and hence differ from bound anaphors and instead resemble free pronominals. I will assume that anaphors can fail to be exempt either because they do not appear in a logophoric context – for instance, by having an inanimate antecedent – or because they appear in a syntactic position that does not allow logophoric anaphors.

First, as already discussed extensively, exempt anaphors do not require local c-command:

- (16a) Kamana_i thought [that James was proud of her_i].
- (16b) Kamana_i thought [that James was proud of the picture of herself_i].
- (16c) *Kamana_i thought [that James was proud of herself_i].
- (16d) *The novel_i profited from the fact [that the pope had been spotted with a copy of itself_i].

Secondly, only exempt anaphors can take split or partial antecedents like pronominals:

- (17a) Kamana_i, who was standing next to James_j, said that they_{i+j} would be late for the party.

9 As for cross-linguistic evidence, logophoric effects have been attested in Icelandic (Maling 1984, Sells 1987, Thráinsson 1991, Sigurjónsdóttir and Hyams 1992), French (Zribi-Hertz 1995, Charnavel and Sportiche 2016, Charnavel 2019), Italian (Napoli 1979), Latin (Clements 1975, Milner 1987), Mandarin Chinese (Anand 2006, Huang et al. 2009) and Japanese (Kuno 1972, Sells 1987), among others.

10 Charnavel (2019) has shown that complementarity with pronominals (e.g. Bouchard 1984, Lebeaux 1984) is not a valid diagnostic for exemptness.

- (17b) Kamana_i, who was standing next to James_j, said that the picture of themselves_{i+j} had won a prize.
- (17c) *Kamana_i, who was standing next to James_j, talked about themselves_{i+j}.
- (17d) *[The blockbuster]_i, which was based on [a best-selling novel]_j, led to numerous imitations of themselves_{i+j}.
- (18a) Kamana_i and James_j believed that she_i was a talented engineer.
- (18b) Kamana_i and James_j liked the picture of herself_i.
- (18c) *Kamana_i and James_j believed in herself_i.
- (18d) *The novel_i and its success_j led to numerous imitations of itself_i.

Thirdly, under the assumption that bound variables must have a sloppy reading in focus constructions and under VP ellipsis (Sag 1976), the availability of strict readings signals co-referential, that is logophoric use, demonstrated here for VP ellipsis:

- (19a) Kamana was proud of her children, and so was Lea.
 Sloppy: Lea was proud of Lea's children.
 Strict: Lea was proud of Kamana's children.
- (19b) Kamana was proud that the picture of herself had won the prize, and so was Lea.
 Sloppy: Lea was proud that the picture of herself had won the prize.
 Strict: Lea was proud that the picture of Kamana had won the prize.
- (19c) Kamana was proud of herself, and so was Lea.
 Sloppy: Lea was proud of herself.
 ??Strict: Lea was proud of Kamana.
- (19d) Kamana's article contains a reference to itself, and so does Lea's.
 Sloppy: Lea's article contains a reference to itself.
 *Strict: Lea's article contains a reference to Kamana's article.

However, the claim that strict readings under VP ellipsis are impossible for bound variables has been contested both empirically and theoretically (e.g. Hestvik 1995, Hardt and Romero 2004, Hardt 2008, Buring 2005), which casts some doubt on this diagnostic. More importantly, German does not have VP ellipsis of the English type, but instead the following two similar constructions (Winkler 2005):

- (20a) Kamana sollte stolz auf sich sein, und Lea sollte es auch.
 Kamana should proud of herself be, and Lea should it too.

Kamana should be proud of herself, and Lea should as well.

(20b) Kamana ist stolz auf sich, und Lea auch.

Kamana is proud of herself, and Lea too.

In (20a), the ellipsis site is represented by a proform which picks up on an already established referent. This might make strict readings more available even for bound variables. (20b), on the other hand, lacks an auxiliary and thus rather patterns with stripping in English, which has been argued to allow for strict readings under different conditions than VP ellipsis (Kennedy and Lidz 2001). I thus conclude that both ellipsis constructions in (20) cannot serve as reliable diagnostics for our purposes.

Fortunately, the same effects are argued to also hold for focus constructions (capitalization signals main sentential stress):

(21a) Only KAMANA was proud of her children.

Sloppy: No one else was proud of her/his children.

Strict: No one else was proud of Kamana's children.

(21b) Only KAMANA was proud that the picture of herself had won the prize.

Sloppy: No one else was proud that the picture of herself/himself had won the prize.

Strict: No one else was proud that the picture of Kamana had won the prize.

(21c) Only KAMANA was proud of herself.

Sloppy: No one else was proud of herself/himself.

?Strict: No one else was proud of Kamana.

(21d) Only KAMANA's article contained references to itself.

Sloppy: No other article contained references to itself.

*Strict: No other article contained references to Kamana's article.

As demonstrated here for (21c), repeated as (22), strict readings should be impossible if the pronoun is bound:

(22) Only KAMANA was proud of herself.

Sloppy: James is not in the set $\{x \mid x \text{ was proud of } x\}$, Lea is not in the set $\{x \mid x \text{ was proud of } x\}$...

?Strict: James is not in the set $\{x \mid x \text{ was proud of Kamana}\}$, Lea is not in the set $\{x \mid x \text{ was proud of Kamana}\}$...

In fact, however, strict readings are moderately acceptable with co-argumental anaphors as in (21c) (Büring 2005), but fully inaccessible with inanimate antecedents as in (21d). This unexpected contrast also holds to some extent for VP ellipsis.

Charnavel and Sportiche (2016) and Charnavel (2019) take this as evidence that co-argumenthood does not per se block exemptness, contra Reinhart and Reuland's (1991, 1993) approach presented above. Instead, they argue that logophoric anaphors are disallowed in certain syntactic position because they compete with morphologically weaker elements, namely clitics in French and unstressed anaphors in English. For reasons of space I cannot do full justice to this proposal here, but will only point out that it does not resolve the issue at hand: If the exempt anaphor in (21c) is ruled out because of the abovementioned morphological competition – which appears to be the case, given (16c) –, the strict reading should be fully unavailable, as in (21d). If it does not compete, the strict reading should be fully available, as in (21b). Either way, the moderate acceptability remains mysterious. I am not aware of any successful solution to this issue.

Since the strict-and-sloppy diagnostic hence poses problems that cannot be solved here, I will not draw on it. I will use the first diagnostic, local c-command, in the experiment presented in chapter 3 as well as in chapter 4, and the second diagnostic, split and partial antecedents, in chapter 4. First, however, we must take a brief look at anaphors in German.

2.3 Logophoric Anaphors in German

So far, we have seen that in various languages, anaphors have been observed to apparently violate Principle A, explained either in terms of logophoricity or of long-distance binding. German anaphors, however, have widely been claimed to always require a local binder (Kiss 2003, 2012, Büring 2005, Cole et. al 2006, Reuland 2006, Platzack 2012, Fischer 2015, Georgi et al. 2019).¹¹ Attested examples such as the following newspaper headline have not been taken as counter-evidence, but as curious outliers:

- (23) ??Der Glaube an sich macht ihn stark: Oliver Kahn.
 The faith in himself makes him strong: Oliver Kahn.
 (Kiss 2003:180, taken from: Süddeutsche Zeitung, 20.06.02)

11 As the only exception to this rule that I am aware of, Gunkel (2003) has proposed that adverbial PP adjuncts in ECM constructions can be long-distance bound, but rules out that they are exempt from binding altogether. Given that binding domains of ECM constructions are particularly controversial, I will not attempt to assess this claim.

However, the claim that logophoricity is absent from German has been based on individual introspective judgments only: So far, there has been no systematic investigation testing whether sentences in which exempt anaphors are predicted to occur, given cross-linguistic evidence, are in fact rejected by speakers.

On the other hand, experimental evidence from a different area has recently lead Georgi et al. (2019) to hypothesize that German anaphors might show some logophoric behaviour after all. Investigating Principle A effects under *wh*-movement to examine reconstruction behaviour, Georgi et al. also tested in situ sentences as a control condition and found curious results:

- (24) Maria erzählt, dass Anna die bronzene Statue von sich gesehen hat.
Maria says that Anna the bronz statue of herself seen has.
Maria says that Anna has seen the bronz statue of herself.

Around 30%¹² of subjects accepted co-reference of *sich* with the matrix subject. These results are puzzling under the assumption that German anaphors must obey Principle A, but fall neatly into place once we assume that the animate subject *Maria* allows the embedded anaphor to be used logophorically. Additionally, the prediction that logophoric anaphors are not licensed in object position is also borne out: Co-reference with the matrix subject was accepted less often when *sich* occurred in object position (less than 10%), but this number increased to around 25% under coordination:

- (25a) Maria erzählt, dass Anna sehr stolz auf sich ist.
Maria says that Anna very proud of herself is.
Maria says that Anna is very proud of herself.
- (25b) Maria erzählt, dass Anna sehr stolz auf sich und die Mannschaften ist.
Maria says that Anna very proud of herself and the teams is.
Maria says that Anna is very proud of herself and the teams.

These findings, which suggest that German might in fact allow for logophoric anaphors, as well as the lack of reliable empirical evidence so far, hence call for a systematic empirical study on logophoricity in German.

12 Subjects were not forced to choose between the potential co-referents but could accept both.

3 Experiment

To gather conclusive evidence whether German anaphors can be used logophorically, I conducted an experiment which I present in the following. While the data seem to offer evidence for logophoric anaphors in German, I will argue in the discussion in chapter 4 that some of the results should be analyzed differently, to some extent calling into question my initial assumption that the effects observed are due to logophoricity.

3.1 Method and Materials

The research question lends itself to two possible types of experiments: an interpretation task that tests whether a logophoric interpretation is available, and an acceptability judgment task that tests whether a given logophoric interpretation is acceptable. I use an acceptability judgment task for three main reasons. First, an interpretation task requires a sentence to have two possible antecedents, one of which locally binds the anaphor and one of which can serve as a logophoric centre licensing an exempt anaphor, as in the sentences used by Georgi et al. (2019). However, testing only sentences in which an alternative, c-commanding antecedent besides the logophoric centre is available limits the range of the experiment considerably. Secondly, logophoric anaphors might be less acceptable when an alternative antecedent intervenes, as observed by Pollard and Sag (1992) for English, which could obscure the effect in question. Thirdly, the study by Georgi et al. (2019) summarised above, although having a different objective, in effect constitutes an interpretation task for logophoricity. Replicating these findings with a different type of experiment strengthens the hypothesis that the effect observed is indeed due to logophoricity.

I test the hypothesis that anaphors in German can be exempt from Principle A iff they occur in a logophoric context and do not occur in object position. Therefore, I predict a sentence containing an anaphor that is not realized as an object and lacks a local¹³ binder to be significantly more acceptable if the anaphor occurs in a logophoric context than if it does not. For the purpose of the experiment, an anaphor is said to occur in a logophoric context if its antecedent is animate and also construed as a perspective holder by means of psych verbs such as *fear*, *be surprised*, *hope*, *be convinced* and *amuse*, *discourage*, *please*, *alarm*. Such verbs were chosen to ensure that the

13 Given the ongoing controversy over the correct definition of binding domains, the experiment will exclusively be concerned with clausal – or tensed TP – domains, without precluding that anaphors might equally be bound in smaller domains (see e.g. Charnavel 2019).

antecedent qualifies as a logophoric centre from whose mental perspective the context is presented.¹⁴ This is not to claim that logophoric anaphors in German will only appear in such contexts, but since the experiment only tests whether exempt anaphors are licensed at all, not under which precise conditions, it uses contexts in which logophoric anaphors are most likely to occur.

More precisely, the experiment tests how the acceptability of sentences is influenced by the interaction of two independent variables, logophoricity and structural condition. The former describes whether an anaphor does or does not occur in a logophoric context. The latter describes the syntactic position of the anaphor as well as its structural relation to its antecedent and takes on four values:

(26a) *Control condition 1 (CC1)*

The anaphor is realized as an object and has a local c-commanding antecedent.

(26b) *Control condition 2 (CC2)*

The anaphor is realized as an object and lacks a local c-commanding antecedent.

(26c) *Test condition 1 (TC1)*

The anaphor is embedded in an argument and has a local, but non-c-commanding antecedent.

(26d) *Test condition 2 (TC2)*

The anaphor is embedded in an argument and has a non-local, but c-commanding antecedent.

We expect sentences to be fully grammatical in CC1 and fully ungrammatical in CC2, regardless of logophoricity. Besides confirming that logophoricity does not influence acceptability in these cases, the results from the control conditions serve as benchmarks for positive and negative ratings of sentences containing anaphors which the results from the test conditions can be compared with. In the test conditions, on the other hand, sentences are predicted to receive significantly higher ratings if they contain a logophoric context. Including two distinct test conditions aims to minimize the risk that the effect, if observed, is caused by some random intervening factor.

Table 1 summarises the resulting 8 variable level combinations:

14 Verbs of saying, which might have generated similar results, were excluded from the experiment: In German, the embedded clause in indirect speech can optionally appear in subjunctive mood (*Konjunktiv I*), the use of which depends to a large extent on the style of speech and varies considerably between speakers. Ratings might hence be influenced by speakers' different judgments on verbal mood in indirect speech, which could distort the results.

	+log	-log
CC1	CC1 +log	CC1 -log
CC2	CC2 +log	CC2 -log
TC1	TC1 +log	TC1 -log
TC2	TC2 +log	TC2 -log

Table 1: Schematic token set

To test these conditions, 16 token sets of 8 sentences each were designed. An exemplary concrete token set is shown here:

(27a) *CC1 +log*

[Der Schauspieler]_i präsentiert sich_i auf dem roten Teppich den
 [The actor]_i presents himself on the red carpet the.DAT
 Fotografen.
 photographers.DAT.
 [The actor]_i presents himself_i to the photographers on the red carpet.

(27b) *CC1 -log*

[Die Statue]_i präsentiert sich_i nach der Restauration dem Betrachter ganz
 [The statue]_i presents itself_i after the restoration the.DAT viewer.DAT completely
 neu.
 new.
 After the restoration [the statue]_i presents itself_i to the viewer in a completely new way.

(27c) *CC2 +log*

Das Bild von [dem Schauspieler]_i gefällt sich_i gut.
 The picture of [the actor]_i pleases himself_i well.
 The picture of [the actor]_i pleases himself_i.

(27d) *CC2 -log*

Das Bild von [der Statue]_i zeigt sich_i nach der Restauration.
 The picture of [the statue]_i shows itself_i after the restoration.
 The picture of [the statue]_i shows itself_i after restoration.

(27e) *TC1 +log*

Das Foto von sich_i gefällt [dem Schauspieler]_i gut.
 The photo of himself_i pleases [the actor]_i well.
 The photo of himself_i pleases [the actor]_i.

(27f) *TC1 –log*

Das Foto von sich_i zeigt [die Statue]_i nach der Restauration.

The picture of itself_i shows [the statue]_i after the restoration.

The picture of itself_i shows [the statue]_i after restoration.

(27g) *TC2 +log*

[Der Schauspieler]_i ist überzeugt, dass das angebliche Foto von sich_i eine

[The actor]_i is convinced that the alleged photo of himself_i a

Fälschung ist.

fake is.

[The actor]_i is convinced that the alleged photo of himself_i is a fake.

(27h) *TC2 –log*

[Die lang verschollene, nun wiedergefundene Statue]_i beweist, dass das angebliche

[The long lost, now rediscovered statue]_i proves that the alleged

Foto von sich_i eine Fälschung sein muss.

photo of itself_i a fake be must.

[The long-lost, now rediscovered statue]_i proves that the alleged photo of itself_i must be a fake.

In CC1, the antecedent appears in subject and the anaphor in object position, and in CC2, the anaphor appears in object position while the antecedent is embedded in the subject. In the test conditions, anaphors are realized as arguments¹⁵ contained in definite¹⁶ NPs: In TC1, the anaphor is embedded in the subject, while its antecedent is in object position. In TC2, the anaphor is embedded in the subject of a complement clause and takes the matrix subject as an antecedent, such that no alternative co-referent occurs between the two. None of the DPs containing the anaphors contained a possessor in specifier position.

To generate a sufficient number of sentences, token sets were designed in such a way as to ensure subjects could see two items, differing across both variable levels, from each token set without noting any duplication. As can be verified for the examples above, CC1 +log can be

15 The argument status of adnominal complements has been questioned e.g. by Grimm and McNally (2013) and Bruening and Al Khalaf (2018). For the present purposes, nothing hinges on this question. Exempt anaphors could also be contained in adjuncts or in coordinated structures, but the conditions were kept as stable as possible to ensure consistent findings.

16 On the positive effect of definiteness on the acceptability of logophoric anaphors in English see Pollard and Sag (1992:273).

combined with TC1 –log, CC1 –log with TC1 +log, CC2 +log with TC2 –log and CC2 –log with TC2 +log.

Additionally, 48 filler items were designed, 16 of which are grammatical, 16 ungrammatical and 16 of intermediate acceptability. Besides distracting from the experiment proper, the fillers also served the purpose of testing whether subjects performed adequately on the acceptability judgment task: Subjects were excluded from the analysis if they rated more than 80% of the grammatical filler items lower than 3 or more than 80% of the ungrammatical filler items higher than 3. All materials can be found in appendix A.

3.2 Procedure and Participants

The set of 128 experimental items in total was divided into four sets of 32 items, each of which contained two sentences from each token set as outlined above. Subjects were randomly assigned one of these four sets of items, such that every item and every variable level combination was seen by roughly equally many subjects. Including the 48 fillers, all participants hence saw 80 sentences.

The experiment was set up using the online behavioural experiment builder Gorilla, and participants completed the task unsupervised at home. After having signed an informed consent form, subjects were first asked to provide information on their age, gender and geographic origin, and received instructions on how to carry out the acceptability judgment task. The experimental items and fillers were then displayed in a pseudorandomized order on the screen. Subjects rated these sentences on a 5 point Likert scale, whose upper and lower end points were labelled *völlig natürlich* (“completely natural”) and *völlig falsch* (“completely false”). Subjects could spend as much time as they wished on each sentence, but could not omit ratings. Finally, subjects were asked to comment on their experience of the experiment, in particular about whether they noticed any patterns or adopted any particular strategies.

Participants were recruited using a non-random snowball sampling approach. Overall, 70 subjects completed the experiment, 2 of which were removed from the analysis due to technical problems, reported in the final question, which prevented them from carrying out the experiment properly. 2 further subjects were removed since they failed to answer 80% of the filler items correctly, as outlined above.¹⁷ Participants had a median age of 31, considerably lower than the

¹⁷ 3 ungrammatical fillers received an unusually high number of wrongly positive ratings (no. 137, 140, 141; see appendix). These sentences contained case and inflection errors reflected in only minor changes of spelling, suggesting that subjects rated these sentences wrongly positively because they simply overlooked the errors. Hence, these filler items were not included in the

median age of the German population, and the sample included more female than male participants (58% to 42%). In terms of geographic origin, participants heavily clustered around the west of Germany, with almost two thirds coming from the federal state of North Rhine-Westphalia. Finally, several participants reported in the final question that they assumed the experiment to be about reflexives, which might have influenced their responses. The full set of demographic data can be found in appendix B.

3.3 Analysis

The data was analysed using the software R. The effect of the independent variables on acceptability rating was computed using a mixed regression model with interaction terms; potential random effects were subject, item and age of subjects. Since the dependent variable is ordinal, a mixed effects ordinal logistic regression was the first choice, computed with the function `clmm()` from the ordinal package. However, it could not be confirmed that the data fulfill the assumptions of the model: Proportionality of odds, which the ordinal logit requires, cannot be verified for mixed models with the usual methods. The assumption was hence tested for a simpler ordinal model without interaction terms and without random effects, using the Brant test – available as the function `brant()` from the brant package – and confirming the results with the graphical method proposed by Harrell (2001:335). The results showed that the proportional odds assumption is violated for the simple additive model. Whether the inclusion of interaction terms and random effects sufficiently alleviates this violation is uncertain. In consequence, results generated by the ordinal model fail to be fully reliable.

This predicament suggests computing a multinomial logistic regression instead, which still correctly captures the categorical output values, but does not make any assumptions about the proportionality of odds. However, as a contingency table showed, the data contains several small and one empty cell, which can make multinomial models unstable (Heinze and Schemper 2002, Allison 2008). Hence, this option had to be rejected as well.

Therefore, a linear mixed model was computed, using the function `lme()` from the nlme package. Linearity and non-multicollinearity are satisfied vacuously, and homoscedasticity of the error could be verified graphically by plotting the distribution of residuals for each level of the independent variables. What appears to raise a problem, leaving aside the ordinal nature of the dependent variable, is the fact that the residuals of the model do not follow a Gaussian distribution, as confirmed by the Shapiro-Wilk test and Q-Q plots. Since the overall distribution of output values is

80% test. All other filler items received much fewer wrong responses.

bimodal, transformations of the dependent variable could not alleviate this violation. However, Gelman and Hill (2007) argue that for mixed models, normality of the error is of little to no importance for estimating the regression line, and in fact advise against testing normality of residuals for such models. Similarly, Jaqmin-Gadda et al. (2007) tested the robustness of linear mixed models with non-Gaussian error and found that inference on fixed effects was not impaired. In sum, the linear model has minor limitations, in particular presupposing a ratio-scaled output variable, but is the most appropriate choice for the present study.

When fitting the linear model, the inclusion of subject as a random effect with random intercept and random slope significantly improved the model. Including item as a random effect did not have a substantial effect on goodness-of-fit, whereas the effect of age was almost as strong as that of subject. Combining several random effects did not lead to significant improvements compared to a single effect, so that in the interest of economy, the latter was preferred. Subject was chosen over age for two main reasons: First, the variable age is unevenly distributed, mostly clustering around two age groups and containing gaps and outliers.¹⁸ It is unclear whether predictions made based on these data are fully reliable. Secondly, subject is preferable to age in the interest of methodological parsimony: Ascribing the variation in the data to variation between subjects is a simpler explanation than ascribing it to the factor age, and since both accounts capture the data equally well, the simpler theory of inter-subject variation is to be preferred.

3.4 Results

Figure 1 visualises the overall distribution, central tendency and dispersion of ratings for the eight variable level combinations, and table 2 presents the results of the regression model. The full set of experimental results can be found in appendix B.

18 Age was additionally coded as a categorical variable with two values, above and below 40 years, since most data points fell either between 20 and 30 years or above 50 years (see figure 2 in appendix B). However, age as a categorical variable improved the model less than as a continuous variable.

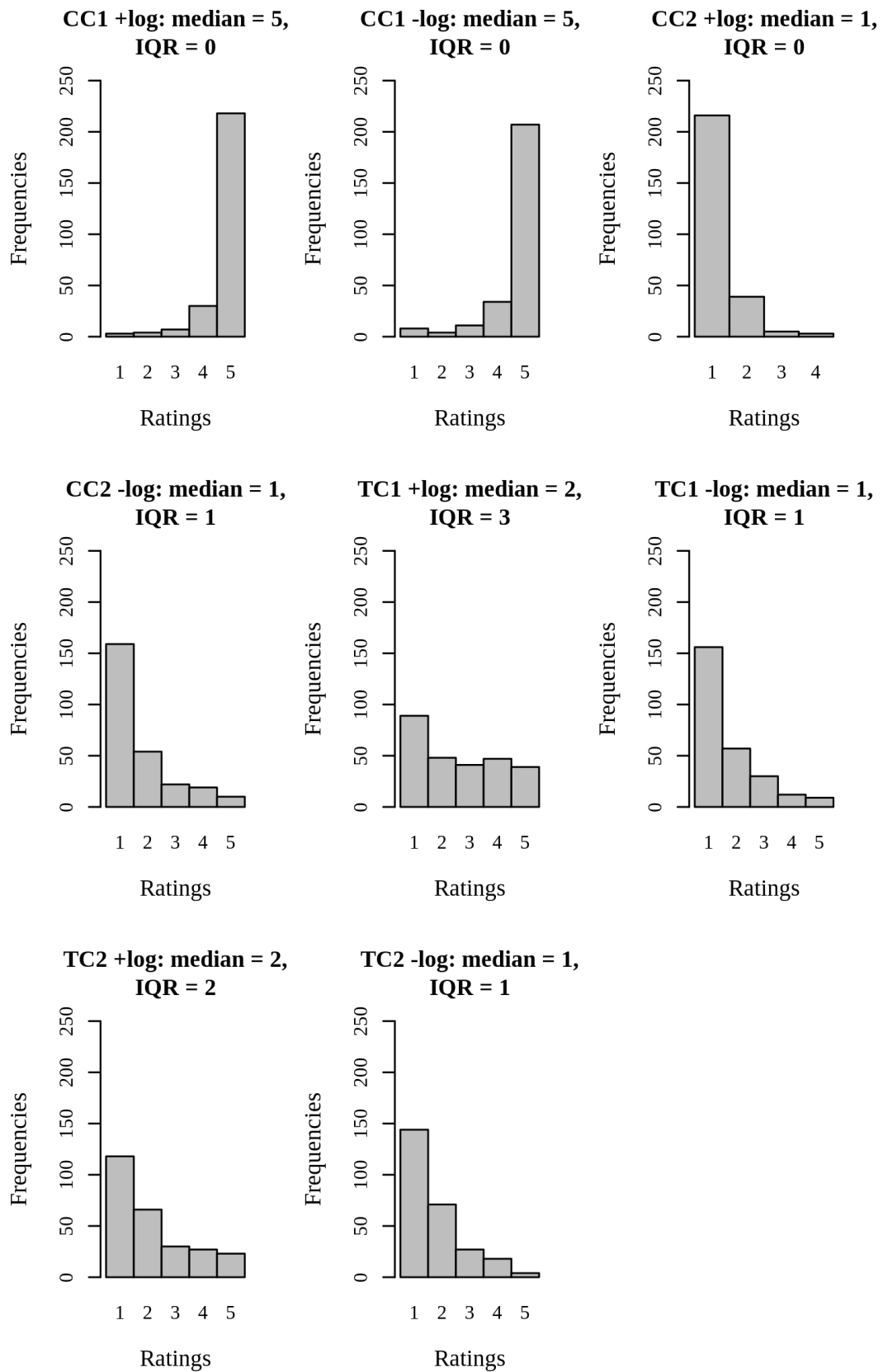


Figure 1: Descriptive statistics

	Value	Std. Error	t-value	p-value
(Intercept)	4.621212	0.06084435	75.95138	0.0000***
+log	0.120357	0.08032511	1.49838	0.1342
CC2	-2.882576	0.08886524	-32.43761	0.0000***
TC1	-2.905303	0.11297145	-25.71714	0.0000***
TC2	-2.882576	0.11396548	-25.29341	0.0000***
+log and CC2	-0.635636	0.11354153	-5.59827	0.0000***
+log and TC1	0.781158	0.11348605	6.88329	0.0000***
+log and TC2	0.273582	0.11348605	2.41071	0.0160*

*Table 2: Fixed effects of linear mixed-effects model fit by REML
(AIC: 5909.853, BIC: 6017.206)*

To summarise the findings, the positive control condition CC1 received strongly positive and the negative control condition CC2 strongly negative responses, as predicted. However, we see a highly significant effect of logophoricity on CC2, such that sentences with a logophoric context are less acceptable than those without. This result is unexpected at first, but can easily be accounted for: Subjects who rated CC2 –log sentences such as (28) as relatively acceptable in all likelihood co-indexed the anaphor with the subject as a whole, not with the intended antecedent embedded in the subject:

- (28) [Das Bild von der Statue_i]_j zeigt sich_{*i/?j} nach der Restauration.
 [The picture of the statue_i]_j shows itself_{*i/?j} after the restoration.
 [The picture of the statue_i]_j shows itself_{*i/?j} after restoration.

Under such a reading, the sentence is both semantically and grammatically flawed, but more acceptable than the reading that involves co-indexation of *sich* with *Statue*, which violates Principle A. Crucially, the same strategy cannot be applied if the sentence contains a logophoric context:

- (29) [Das Bild von dem Schauspieler_i]_j gefällt sich_{*i/*j} gut.
 [The picture of the actor_i]_j pleases himself_{*i/itself}_{*j} well.
 [The picture of the actor_i]_j pleases himself_{*i/itself}_{*j}.

Since the CC2 +log sentences contain EO psych verbs requiring an animate object, the subject *the picture of the actor*, being inanimate, is not available as a potential antecedent. In CC2 +log, subjects could thus not avoid the interpretation that entails a Principle A effect, resulting in

univocally negative responses. The slightly higher ratings for CC2 –log are due to experimental error, namely subjects interpreting the sentences in such a way as to avoid a Principle A violation. Only CC2 +log can hence be taken as the negative baseline which the test conditions can be compared with.

Furthermore, the results also show a highly significant effect of logophoricity on TC1, such that anaphors were more acceptable in the logophoric than in the non-logophoric condition. This appears to support the hypothesis that anaphors in German can in fact be used logophorically. TC1 +log is nevertheless not as acceptable as the positive control condition CC1: Rather, the results are fairly evenly spread out across the five rating levels, also reflected in an IQR of 3, higher than in every other variable level combination. Crucially, the model improved significantly when adding subject as a random effect, whereas accounting for inter-item variation had little impact on the model's goodness-of-fit. This demonstrates that the high variation in TC1 +log is not due to the fact that only some instances of such anaphors are acceptable, but that subjects accept these sentences to different degrees. Similarly, logophoricity has a significant effect on TC2, comparable to TC1, but considerably weaker.

Before turning to the discussion, I will now recapitulate the limitations of the study. First, due to the sampling approach used, the sample is biased in terms of its demographic make-up. The results can hence only be generalized with caution; in particular, it cannot be excluded that the effect observed differs across local varieties of German. Secondly, as the final question showed, the experiment was not fully blinded. The reason for this is that due to the limited sample size and the large number of variable level combinations, the experiment necessarily comprised a large number of experimental items per subject. A filler-item proportion of 3:1, 4:1 or even larger, which could have prevented the unblinding, would have resulted in a very lengthy experiment, thus entailing the risk that participants lose focus or even abandon the task. It is possible that the fact that subjects were aware of the effect under observation influenced their responses. Finally, the linear mixed model used for the data analysis wrongly presupposes ratio-scaled output data and normally distributed residuals, the impact of which, however, has been argued to be rather negligible.

4 Discussion

The preceding chapter has shown that some, although not all speakers of German accept unbound anaphors in logophoric contexts, which seems to indicate that *sich* can be used logophorically. However, in this chapter I will discuss two challenges to this conclusion: In 4.1, I will show that the results from TC1 can be explained as the result of backward binding. In 4.2, I will discuss whether

the results from TC2 should be analysed as instances of long-distance binding, but will eventually adopt a more complex hypothesis, presented in 4.3.

4.1 Logophoricity vs Backward Binding

The experiment has confirmed the following contrast in acceptability:

- (30a) Das Foto von sich_i gefällt [dem Schauspieler]_i gut.
The photo of himself_i pleases [the actor]_i well.
The photo of himself_i pleases [the actor]_i.
- (30b) ??Das Foto von sich_i zeigt [die Statue]_i nach der Restauration.
The photo of itself_i shows [the statue]_i after the restoration.
The photo of itself_i shows [the statue]_i after restoration.

So far, we have assumed that (30a), *prima facie* at odds with Principle A, is acceptable because the animate antecedent can license a logophoric anaphor, unlike the inanimate antecedent in (30b).

However, this wrongly predicts (31) to be acceptable:

- (31) ??Das Foto von sich_i zeigt [den Schauspieler]_i auf dem roten Teppich.
The photo of himself_i shows [the actor]_i on the red carpet.

Crucially, the TC1 +log sentences such as (30a) all involved experiencer object psych verbs, which was motivated by the attempt to ensure that the antecedent constitutes a perspective-holder. We might hence attribute the unacceptability of (31) to the fact that the antecedent does not qualify as a logophoric centre. However, as can be observed in (32), EO psych verbs also allow quantifier binding into the subject:¹⁹

- (32) Das Foto von seiner_i Mutter gefällt [jedem Jungen]_i gut.
The photo of his_i mother pleases [every boy]_i well.
The photo of his_i mother pleases [every boy]_i.

In constructions without experiencer object, binding by the object into the subject is impossible:

- (33) ??Das Buch über seine_i Mutter beschreibt [jeden Jungen]_i als einen guten Sohn.
The book about his_i mother describes [every boy]_i as a good son.

¹⁹ I owe this observation to Susi Wurmbrand (pers. comm.).

Since we see a contrast between (30a) and (31), but not between (30b) and (31), we can infer that the crucial factor responsible for the acceptability of (30a) is not the animacy of the antecedent, but the EO psych verb. Hence, the fact that all +log sentences involved psych verbs, aimed at creating a logophoric context, obscured the true cause of the contrast between the +log and –log conditions.

Such apparent Principle A violations triggered by EO psych verbs, for which I will adopt the term backward binding, have long been observed for various languages:

(34a) Pictures of each other_i annoy [the politicians]_i.

(34b) Stories about herself_i generally please Mary_i.

(34c) Each other_i's health worried [the students]_i.

(34d) *Each other_i's friends murdered [the men]_i.

(Pesetsky 1987:127)

(35a) Questi pettegolezzi su di sé_i preoccupano Gianni_i più di ogni altra
 These gossips about of himself_i worry Gianni_i more than every other
 cosa.
 thing.

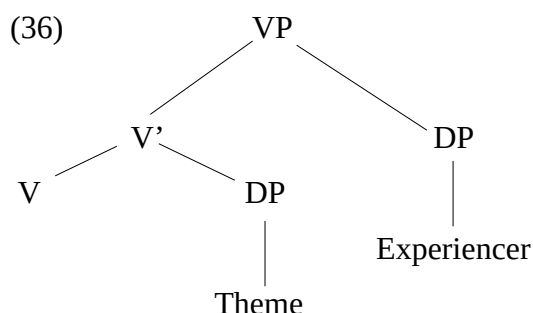
These gossips about himself_i worry Gianni_i more than anything else.

(35b) *Questi pettegolezzi su di sé_i descrivono Gianni_i meglio di ogni
 These gossips about of himself_i describe Gianni_i better than every
 biografia ufficiale.
 biography official.

These gossips about himself_i describe Gianni_i better than any official biography.

(Belletti and Rizzi 1988:312)

To capture these effects, Belletti and Rizzi (1988) have argued that sentences containing EO psych verbs show peculiar binding effects due to their underlying syntactic structure. They propose that experiencer objects are merged in a position c-commanding the theme:



As a result, in (35a), but not in (35b), the antecedent underlyingly c-commands the anaphor, which results in the observed contrast in grammaticality. To derive the correct spell-out, Belletti and Rizzi argue that EO psych verbs have unaccusative properties and fail to assign case to the theme,²⁰ which hence A-moves to the subject position.

Although certain aspects of Belletti and Rizzi's analysis are controversial (see Temme 2018 for a recent overview over various points of criticism and alternative approaches), there is little doubt that EO psych verbs, besides other curious properties, trigger backward binding due to their θ -grid, independently from any logophoric effects. Evidence for backward binding in German in particular comes from Temme (2018), who confirmed experimentally that quantified experiencer objects can bind possessives embedded in the subject, resulting in the following contrast:

- (37a) ?Hin und wieder können die Meinungen seiner_i Schwester jeden_i verwundern.
 Now and then can the opinions his.GEN_i sister.GEN everyone_i surprise.
 Now and then the opinions of his_i sister can surprise everyone_i.
- (37b) *Hin und wieder würden die Schulkameraden seiner_i Schwester jeden_i hänseln.
 Now and then would the classmates his.GEN_i sister.GEN everyone_i tease.
 Now and then the classmates of his_i sister would tease everyone_i.
 (Temme 2018:74)

If we assume that the experiencer is merged higher than the theme, these effects are as expected.²¹

Therefore, we must adjust our interpretation of the experimental results from TC1: Instead of offering evidence for logophoric anaphors, they demonstrate backward binding of anaphors in German. While unexpected, these findings are nevertheless of interest, given that such constructions have long been deemed unacceptable in the literature. On the other hand, non-c-commanded anaphors in the matrix clause that cannot be rescued by backward binding lead to ungrammaticality:

- (38a) ??Die neue Biografie über sich_i stellt [den Schauspieler]_i im besten Licht dar.
 The new biography about himself_i presents [the actor]_i in.the best light dar.
 PRE.

20 This forces Belletti and Rizzi to postulate that the experiencer is assigned lexical case.

21 What is unexpected under this analysis is that backward binding is unacceptable for many speakers of German, as confirmed both by Temme and the present study. Given that these constructions are assumed to comply with Principle A, Temme suggests to attribute their marginal acceptability to pragmatic factors, in particular high processing costs.

The new biography about himself_i presents [the actor]_i in the best light.

(38b) ??Die Pressekampagne gegen sich_i ruiniert [den Schauspieler]_i finanziell.

The press.campaign against himself_i ruins [the actor]_i financially.

The press campaign against himself_i ruins [the actor]_i financially.

(38c) ??Das Porträt von sich_i ähnelt [dem Schauspieler]_i bis ins kleinste Detail.

The portrait of himself_i resembles [the actor]_i to in.the smallest detail.

The portrait of himself_i resembles [the actor]_i down to the smallest detail.

(38d) *Trotz der Kritik an sich_i interviewt die Reporterin [den Schauspieler]_i.

Despite the criticism of himself_i interviews the reporter [the actor]_i.

Despite the criticism of himself_i, the reporter interviews [the actor]_i.

(38e) *Trotz der Kritik an sich_i druckt die Zeitung ein Interview mit [dem

Despite the criticism of himself_i prints the newspaper an interview with [the Schauspieler]_i.

actor]_i.

Despite the criticism of himself_i, the newspaper prints an interview with [the actor]_i.

A curious exception to this rule are sentences such as the following, acceptable both in English and in German, which Charnavel (2019) attributes to logophoricity:

(39) Anonymous posts about herself_i on the internet hurt Lucy_i's feelings.

(Charnavel 2019:90)

(40a) Die abschätzigen Kommentare über sich_i verletzen Lucys_i Gefühle.

The derogatory comments about herself_i hurt Lucy_i's feelings.

(40b) Das vorteilhafte Bild von sich_i schmeichelte Lucys_i Eitelkeit.

The favourable picture of herself_i flattered Lucy_i's vanity.

Even if we assume the experiencer object to be generated high, the antecedents in SpecDP do not c-command the anaphors. However, although syntactically *Lucy* is embedded in the object, semantically there is a clear sense in which she is in fact the experiencer. Interestingly, these sentences also involve idioms. Also considering that in all other constructions, logophoric anaphors in the matrix clause seem to be disallowed in German, it appears that the sentences in (40) should rather be analysed as peculiar instances of backward binding than of logophoricity. If this is correct, this raises further questions about the underlying syntax and semantics of such configurations, which also has a bearing on their analysis in other languages.

To conclude, while the sentences in (40) warrant further investigation, we have seen that the results from TC1 are due to backward binding, not to logophoricity. TC2, on the other hand, will be discussed in the next chapter.

4.2 Logophoricity vs Long-Distance Binding

The experimental results from TC2 showed a contrast between the following sentences:

- (41a) [Der Schauspieler]_i ist überzeugt, dass das angebliche Foto von sich_i eine Fälschung ist.
 [The actor]_i is convinced that the alleged photo of himself_i a fake is.
 [The actor]_i is convinced that the alleged photo of himself_i is a fake.
- (41b) ??[Die lang verschollene, nun wiedergefundene Statue]_i beweist, dass das angebliche Foto von sich_i eine Fälschung sein muss.
 [The long lost, now rediscovered statue]_i proves that the alleged photo of itself_i a fake be must.
 [The long-lost, now rediscovered statue]_i proves that the alleged photo of itself_i must be a fake.

As in TC1 +log, the experimental items in the TC2 +log condition included psych verbs only, involving experiencer subjects. Changing the type of verb seems to have no major impact on grammaticality in these cases:

- (42a) Hamid_i beteuerte, dass das angebliche Foto von sich_i eine Fälschung war.
 Hamid_i affirmed that the alleged photo of himself_i a fake was.
 Hamid_i affirmed that the alleged photo of himself_i was a fake.
- (42b) Hamid_i bewies, dass das angebliche Foto von sich_i eine Fälschung war.
 Hamid_i proved that the alleged photo of himself_i a fake was.
 Hamid_i proved that the alleged photo of himself_i was a fake.
- (42c) Hamid_i profitierte davon, dass das Foto von sich_i in allen Zeitungen erschien.
 Hamid_i profited from this that the photo of himself_i in all newspapers appeared.
 Hamid_i profited from the fact that the photo of himself_i appeared in all newspapers.
- (42d) Hamid_i verdient es, dass das Buch über sich_i so schlechte Kritiken erhalten hat.
 Hamid_i deserves it that the book about himself_i such bad reviews received has.

Hamid_i deserves it that the book about himself_i has received such bad reviews.

There is a slight sense in which non-local anaphors seem to be more felicitous in embedded sentences which are explicitly presented from the perspective of the antecedent by means of ES psych verbs or verbs of saying. While this effect requires further experimental confirmation, it would be in line with the hypothesis that it is the logophoric context which licenses non-local anaphors for (some) speakers of German.

However, as I will argue now, additional data might force us to reconsider this conclusion. Let us first look at the following examples from Belletti and Rizzi (1988):

- (43a) Chiunque dubiti della propria_i buona fede preoccupa Gianni_i.
Whoever doubts of.the own_i good faith worries Gianni_i.
Whoever doubts of [his own]_i good faith worries Gianni_i.
- (43b) *Gianni_i preoccupa chiunque dubiti della propria_i buona fede.
Gianni_i worries whoever doubts of.the own_i good faith.
Gianni_i worries whoever doubts of [his own]_i good faith.
- (43c) *Coloro che vogliono sostenere la propria_i candidatura temono Gianni_i.
Those who want.to support the own_i candidature fear Gianni_i.
Those who want to support [his own]_i candidature fear Gianni_i.
- (43d) Gianni_i teme coloro che vogliono sostenere la propria_i candidatura.
Gianni_i fears those who want.to support the own_i candidature.
Gianni_i fears those who want to support [his own]_i candidature.
(Belletti and Rizzi 1988:321–322)

These sentences all contain a non-local anaphor in a relative clause with an antecedent in the matrix clause, which appears to negatively affect grammaticality in (43b) and (43c), but not in (43a) and (43d). We can attribute the contrast between (43c) and (43d) to the fact that in (43d), but not in (43c), the antecedent c-commands the anaphor. The same effect holds for (43a) and (43b) if we assume that the experiencer object underlyingly c-commands the theme, as proposed by Belletti and Rizzi and discussed above.²² The seemingly paradox judgments in (43) thus follow from Belletti and Rizzi's analysis of psych verbs.

22 Belletti and Rizzi (1988) assume that Principle A can be satisfied at any stage of the derivation.

Ruling out that in (43b) *Gianni* binds the anaphor at S-structure requires them to make the somewhat ad-hoc stipulation that long-distance bound anaphors need to be bound from a θ -position, which excludes SpecIP.

Similar examples can be construed for German, using reflexives instead of possessives:

- (44a) Gianni_i hat Angst vor dem Reporter, der einen Artikel über sich_i schreibt.
 Gianni_i has fear of the reporter who an article about himself_i writes.
 Gianni_i is afraid of the reporter who writes an article about himself_i.
- (44b) *Der Reporter, der einen Artikel über sich_i schreibt, hat Angst vor Gianni_i.
 The reporter who an article about himself_i writes has fear of Gianni_i.
 The reporter who writes an article about himself_i is afraid of Gianni_i.
- (44c) *Gianni_i beunruhigt den Reporter, der einen Artikel über sich_i schreibt.
 Gianni_i worries the reporter who an article about himself_i writes.
 Gianni_i worries the reporter who writes an article about himself_i.
- (44d) Der Reporter, der einen Artikel über sich_i schreibt, beunruhigt Gianni_i.
 The reporter who an article about himself_i writes worries Gianni_i.
 The reporter who writes an article about himself_i worries Gianni_i.

We can apply Belletti and Rizzi's analysis of the Italian data to German and argue that only in (44a) and (44d), the non-local anaphor is (underlyingly) c-commanded by its antecedent and therefore acceptable, by adopting the view that experiencer objects are generated high. However, this entails that anaphors with non-local antecedents require c-command. This is incompatible with our assumption that non-local anaphors in German are licensed by logophoricity, since the latter implies a co-referential use which should not rely on c-command. Rather, this fact suggests that the anaphors in question are instances of long-distance binding: As we have seen in chapter 2, long-distance binding approaches assume that some anaphors can – at least on the surface²³ – take a non-local antecedent, but are not exempt from binding altogether, hence still requiring c-command.

First, let us evaluate whether this hypothesis is compatible with the experimental findings. Since in TC2, the antecedent is realized as the matrix subject while the anaphor is contained in the embedded subject, c-command is always satisfied. At the same time, sentences received significantly higher ratings if containing animate antecedents, which strongly supports a logophoricity instead of a long-distance binding approach. However, despite clearly reaching significance level, we cannot preclude that this effect was due to experimental error and might not be replicable. For instance, sentences with inanimate antecedents and clausal complements, as were used for the TC2 –log condition, are rare, difficult to construct and might have been more likely to

²³ While pursuing this hypothesis, I do not make any assumptions about the details of the long-distance binding analysis, e.g. whether the anaphor undergoes movement or can simply be bound in a larger domain.

be rejected on independent grounds. Hence, the experimental results alone, while backing up a logophoricity account, cannot definitively rule out long-distance binding.

But there is an alternative analysis of the sentences in (44) which so far we have taken as evidence for long-distance binding: Only in (44a) and (44d), the antecedent *Gianni* qualifies as a logophoric centre, whereas in (44b) and (44c), the embedded content is presented from the perspective of the reporter. Under this view, the non-local antecedents in (44b) and (44c) are not degraded because they do not c-command the anaphor, but because they are not logophoric centres.²⁴

On the other hand, long-distance binding makes the correct predictions for (45):

(45a) *Ich habe von Hamid_i gehört, dass das Foto von sich_i eine Fälschung sei.

I have from Hamid_i heard that the photo of himself_i a fake is.SUBJ.

I have heard from Hamid_i that the photo of himself_i is a fake.

(45b) *Hamid_i zufolge ist es unbestreitbar, dass das Foto von sich_i eine Fälschung ist.

Hamid_i according.to is it undeniable that the photo of himself_i a fake is.

According to Hamid_i, it is undeniable that the photo of himself_i is a fake.

It is unclear why these antecedents should not qualify as logophoric centres. At the same time, the antecedents, embedded in PPs, cannot c-command the anaphors, which accounts for the ungrammaticality.²⁵

24 This analysis has the positive side effect that we do not need to devise an additional constraint ruling out binding from SpecIP (see footnote 23).

25 Quantifier binding is possible out of such PPs, resembling inverse linking (see May and Bale 2006 for an overview over the latter):

(i) Ich habe von [jedem Jungen]_i gehört, dass das Foto von seiner_i Mutter eine Fälschung

I have from [every boy]_i heard that the photo of his_i mother a fake

ist.

is.

I have heard from [every boy]_i that the photo of his_i mother is a fake.

(ii) [Jedem Jungen]_i zufolge ist es unbestreitbar, dass das Foto von seiner_i Mutter

[Every boy]_i according.to is it undeniable that the photo of his_i mother

eine Fälschung ist.

a fake is.

According to [every boy]_i, it is undeniable that the photo of his_i mother is a fake.

Counter-evidence for the long-distance binding approach comes from (46):

- (46a) *Ich habe Hamid_i erzählt, dass das angebliche Foto von sich_i eine
I have Hamid.DAT_i told that the alleged photo of himself_i a
Fälschung ist.
fake is.
I told Hamid that the alleged photo of himself is a fake.
- (46b) Die Reaktion seiner Freunde zeigte Hamid_i, dass das Bild von sich_i in der
The reaction his.GEN friends.GEN showed Hamid_i that the picture of himself in the
Tat eine Fälschung war.
fact a fake was.
The reaction of his friends showed Hamid_i that the picture of himself_i was in fact a fake.

In (46a), the antecedent does c-command the anaphor but the sentence is nevertheless unacceptable. This could be accounted for by postulating that non-local anaphors in German are obligatorily subject-oriented, as has been claimed for long-distance binding in several languages (e.g. Thráinsson 1991, Reinhart and Reuland 1991, 1993, Sigurjónsdóttir and Hyams 1992, Cole et al. 2006; see also for a cross-linguistic overview Buring 2005:59). But this makes the wrong predictions for (46b), in which the anaphor is licensed although the antecedent is not the subject. From a logophoric perspective, on the other hand, the antecedent qualifies as a logophoric centre in (46b), but not in (46a), yielding the correct grammaticality effects.

On the other hand, anaphors are generally unacceptable in German if they lack an antecedent in the same sentence altogether, while English allows this under certain circumstances:

- (47a) *Kim_i klopfte und trat ein. Abgesehen von sich_i war niemand sonst im Raum.
Kim_i knocked and stepped in. Apart from herself_i was nobody else in the room.
Kim_i knocked and entered. Apart from herself_i, nobody else was in the room.
- (47b) *Louis_i war nervös. Würde Khoa den Gerüchten über sich_i glauben?
Louis_i was nervous. Would Khoa the rumours about himself_i believe?
Louis_i was nervous. Would Khoa believe the rumours about himself_i?
- (47c) *Martha_i beschloss, ihren Ehemann zu verlassen. Die Anschuldigungen gegen sich_i

Although these phenomena are not well understood, I take it that they are due to the quantificational nature of the binder and hence do not affect the analysis of anaphor binding in (42).

Martha_i decided her husband to leave. The accusations against herself_i waren unerträglich geworden.
 were intolerable become.
 Martha_i decided to leave her husband. The accusations against herself_i had become intolerable.

- (48) Mary_i was extremely upset. That picture of herself_i on the front page of the Times would circulate all over the world.
 (Pollard and Sag 1992:268)

This follows in a straightforward fashion from a c-command requirement on long-distance bound anaphors, whereas a theory based on logophoricity would need to devise an additional rule that disallows such sentences for German.

Equally, logophoric anaphors, by virtue of having a co-referential use, should allow for split and partial antecedents, which is clearly not the case:

- (49a) *Der Minister_i, der neben der Präsidentin_j stand, behauptete, dass die Kritik an
 The minister_i who near the president_j stood claimed that the criticism of
 sich_{i+j} ungerechtfertigt sei.
 themselves_{i+j} unjustified is.SUBJ.
 The minister_i, who was standing near the president_j, claimed that the criticism of
 themselves_{i+j} was unjustified.
- (49b) *Der Minister_i und die Präsidentin_j waren beide der Meinung, dass die Kritik
 The minister_i and the president_j were both the.GEN opinion.GEN that the criticism
 an sich_i ungerechtfertigt war.
 of himself_i unjustified was.
 The minister_i and the president_j were both of the opinion that the criticism of himself_i
 was unjustified.

To sum up, we asked whether non-local anaphors, which are acceptable for some speakers of German under certain circumstances, are long-distance bound or have a logophoric interpretation. However, neither logophoricity nor long-distance binding seem to be able to account for the full set of data. Rather, we can derive all the effects we have seen by combining the two approaches, stating that non-local anaphors in German must both be c-commanded and occur in a logophoric context. Such a principle is obviously at odds with standard assumptions about binding and co-reference:

Logophoric anaphors, having co-referential properties, should be free from any constraint on c-command. Hence, if we accept the data presented here, this forces us to question the disjunction between co-reference and binding as assumed so far.

4.3 Syntactic vs Semantic Binding

To account for the facts outlined, I will now develop a simple theory based on the distinction between syntactic and semantic binding. I call an anaphor syntactically bound iff it is c-commanded by a co-indexed DP, and semantically bound iff its interpretation co-varies with the interpretation of its antecedent.²⁶ Based on these notions, I propose the following constraint on German anaphors:

(50) *Constraint on anaphor binding in German (first version)*

- i. Anaphors must be syntactically bound.
- ii. Anaphors must either be semantically bound or co-refer with a relevant logophoric centre.

I define a relevant logophoric centre as a DP designating one or several individuals from whose mental perspective the context containing the anaphor is presented.

Since anaphors must be co-indexed with their syntactic binder, it appears that the latter must be identical to the anaphor's antecedent, meaning the DP from which the anaphor receives a semantic interpretation. For instance, the anaphor in (51), sharing a referent with *Hamid*, cannot be syntactically bound by *Lea* because they are not co-indexed, and hence violates the constraint:

- (51) *Hamid_i zufolge mag Lea_j das Bild von sich_i.
 Hamid_i according.to likes Lea_j the picture of himself_i.
 According to Hamid_i, Lea_j likes the picture of himself_i.

I will thus assume that anaphors must always be syntactically bound by and receive an interpretation from the same DP. Why such an identity should hold will be discussed at the end of this chapter.

But first we must address an obvious problem: The above constraint is too permissive since it does not specify which anaphors require semantic binding and which can be used logophorically. As a first attempt, we can draw on a rudimentary notion of locality which will be revised later:

²⁶ This is not to claim that the interpretation of the antecedent must in fact vary.

(52) *Constraint on anaphor binding in German (second version)*

- i. Anaphors must be syntactically bound.
- ii. Anaphors must be either semantically bound or co-refer with a relevant logophoric centre.
- iii. Semantically bound anaphors must be bound in a local domain.

As before, I will simply regard the clause as the relevant local domain. The constraint now correctly predicts that non-local anaphors are only acceptable in logophoric conditions, as confirmed experimentally in the TC2 condition, and that anaphors lacking a c-commanding antecedent result in ungrammaticality even if they are used logophorically, as shown in chapters 4.1 and 4.2. A more intricate question is whether the constraint also entails the impossibility of split and partial antecedents:

- (53a) *Der Minister_i und die Präsidentin_j waren beide der Meinung, dass die Kritik
The minister_i and the president_j were both the.GEN opinion.GEN that the criticism
an sich_i ungerechtfertigt war.
of himself_i unjustified was.
The minister_i and the president_j were both of the opinion that the criticism of himself_i
was unjustified.
- (53b) *Der Minister_i, der neben der Präsidentin_j stand, behauptete, dass die Kritik an
The minister_i who near the president_j stood claimed that the criticism of
sich_{i+j} ungerechtfertigt war.
themselves_{i+j} unjustified was
The minister_i, who was standing near the president_j, claimed that the criticism of
themselves_{i+j} was unjustified.
- (53c) *Der Minister_i sagte der Präsidentin_j, dass die Kritik an sich_{i+j}
The minister_i told the.DAT president.DAT_j that the criticism of themselves_{i+j}
ungerechtfertigt war.
unjustified was.
The minister_i told the president_j that the criticism of themselves_{i+j} was unjustified.

Generally, the ban on non-exhaustive anaphor binding could either follow from the requirement on c-command or be an independent property of anaphors. Lack of c-command straightforwardly rules out partial antecedents as in (53a) and the split antecedents in (53b). In (53c), both *der Minister* and *die Präsidentin* do c-command the anaphor, which is, however, irrelevant since neither of them is

the anaphor's antecedent. The actual antecedent is the conjunct of both, which does not appear as such in the sentence and hence also fails to c-command the anaphor. Therefore, the impossibility of split and partial antecedents is a mere consequence of the fact that anaphors require a c-commanding antecedent and therefore follows from the constraint.

While the constraint hence covers a considerable range of data, section (iii) which defines in which contexts semantically bound anaphors are licensed is still in need of fine-tuning. So far, I have imposed a simple locality constraint on semantic binding. This can easily be shown to overgenerate:

- (54) *Kamana_i glaubt, dass James sich_i mag.
Kamana_i thinks that James herself_i likes.
Kamana_i thinks that James likes herself_i.

So far, nothing in the constraint prevents *sich* in (54) from co-referring with the logophoric centre *Kamana*. To amend this, we can replace locality by co-argumenthood:

- (55) *Constraint on anaphor binding in German (third version)*
i. Anaphors must be syntactically bound.
ii. Anaphors must be either semantically bound or co-refer with a relevant logophoric centre.
iii. Anaphors must be semantically bound iff they are syntactically bound by a co-argument.

This captures the previous examples and also makes another apparently correct prediction:

- (56a) Kamana mag das Bild von sich.
Kamana likes the picture of herself.
(56b) ??Das Museum enthält ein Bild von sich.
The museum contains a picture of itself.

The anaphor in (56b), having an inanimate antecedent, is degraded,²⁷ indicating that the anaphor in (56a) is used logophorically. This is evidence that semantic binding is indeed sensitive to co-argumenthood, and not only to locality. On the other hand, (57) is unexpected:

(57) **Kamana_i glaubt, dass James eine Schlange neben sich_i gesehen hat.*²⁸

Kamana_i thinks that James a snake near herself_i seen has.

Kamana_i thinks that James has seen a snake near herself_i.

The anaphor fails to co-refer with the logophoric centre *Kamana*, suggesting that it must be semantically bound by the embedded subject. However, this does not follow from the constraint since the anaphor is not co-argumental. Alternatively, we could assume that the anaphor in (57) does not per se require semantic binding, but cannot be used logophorically, the reasons for which are equally unclear. Therefore, an approach based on co-argumenthood does not seem to account for the full set of data either.

To give a brief interim summary, I argued that anaphors, while always syntactically bound, can be assigned an interpretation either through logophoric co-reference or through semantic binding. I then asked under which conditions which interpretative mechanism applies, that is, which syntactic positions host semantically bound or logophoric anaphors. Empirically, this question requires more, especially more experimental data and hence cannot be solved here conclusively. Generally, I take it that it is semantic binding which requires special syntactic licensing – like locality or co-argumenthood – and that logophoricity, while constrained pragmatically, is the default. This is in

27 Interestingly, the pronominal in (i) is equally rejected by speakers, while (ii) and (iii), containing the intensifier *selbst*, are both approved by some, but not all:

(i) **Das Museum enthält ein Bild von ihm.*

The museum contains a picture of it.

(ii) ?*Das Museum enthält ein Bild von sich selbst.*

The museum contains a picture of itself self.

(iii) ?*Das Museum enthält ein Bild von ihm selbst.*

The museum contains a picture of it self.

28 Non-local anaphors in adnominal PP adjuncts are much more acceptable:

(i) *Kamana_i glaubt, dass die Schlange neben sich_i James gebissen hat.*

Kamana_i thinks that the snake near herself_i James bitten has.

Kamana_i thinks that the snake near herself_i bit James.

line with the economy principle on pronoun interpretation introduced by Reinhart (1983) and Grodzinsky and Reinhart (1993) and elaborated in Reuland (2001), which states that an interpretation based on syntactic processes, namely binding, is preferred when possible. The alternative interpretative mechanism, namely co-reference, can hence only apply when binding is blocked. Although designed to govern the choice between binding and co-reference, we can apply this principle to the distinction between semantic binding and co-reference instead and reformulate the constraint in more general terms:

(58) *Constraint on anaphor binding in German (final version)*

- i. Anaphors must be syntactically bound.
- ii. Anaphors must be semantically bound if possible. Iff semantic binding is blocked, anaphors must co-refer with a relevant logophoric centre.

I thus assume that anaphors do not require semantic binding qua anaphors, but by virtue of appearing in certain configurations. Fleshing out the constraint empirically is a task to be tackled in future research.

Finally, we face the question why all anaphors in German, even if used logophorically, must be syntactically bound by their antecedent. While there is broad consensus that semantic binding requires c-command (e.g. Heim and Kratzer 1998), there is no clear semantic reason why certain cases of co-reference should equally do so. I will sketch out an only tentative solution which might prove a fruitful starting point for further inquiry. Drawing on a theory of agreement as valuation through feature sharing (e.g. Pesetsky and Torrego 2007), Rooryck and Vanden Wyngaerd (2011) have argued that anaphors enter the derivation with interpretable but unvalued ϕ -features. These must subsequently be valued through an AGREE relation with the c-commanding antecedent (see also Reuland 2001, Heinat 2009, Kratzer 2009 and Wurmbrand 2017 for similar proposals).²⁹ We might thus argue that German anaphors, regardless of how they are assigned an interpretation, must be c-commanded by their antecedent in order to receive their ϕ -features from the latter. This also provides a natural explanation of the fact that syntactic binder and semantic antecedent need to be identical.

However, while the hypothesis that the ϕ -features of anaphors are valued via AGREE is not uncommon, the view that would have to be adopted here is new in a fundamental respect: The

²⁹ All these analyses face the obvious challenge that under a standard implementation of AGREE, the unvalued feature acts as a probe and finds a goal in its c-commanding domain, whereas anaphors are c-commanded by their antecedent, hence exhibiting the reverse structural relation. I refer the reader to the cited works for various solutions to this problem.

central claim of the proposals referenced above is that anaphors need to agree with their antecedent in order to enter into a binding relation.³⁰ Accordingly, Rooryck and Vanden Wyngaerd (2011:145–149) claim that co-referential, that is logophoric anaphors enter the derivation already lexically valued and do not undergo AGREE. In contrast, to account for the effects presented above, I must argue that both semantically bound and logophoric anaphors need to agree with their antecedent and thus require c-command. While this is not wholly implausible, it raises an intricate question: If agreement is not required for binding, then we need an account of why anaphors need to have their ϕ -features valued in the first place, as opposed to, for instance, free pronominals.

Moreover, it is unclear why not all languages require their logophoric anaphors to agree and thus be c-commanded. Further research on whether there are other languages that follow the German pattern – and which – would be highly informative. A guiding hypothesis might be that logophoric anaphors only require c-command if their overt morphology is defective, thus being in need of valuation. This would capture, for instance, the contrast between English and German.

I conclude from this brief discussion that linking the c-command requirement on logophoric anaphors in German to agreement might be fruitful, but requires more investigation. Ultimately, we are still lacking a full picture of the relation between agreement and binding, and between agreement and semantic interpretation in general.

5 Conclusion

The question we set out to answer at the beginning of this study was whether German anaphors can have a logophoric use. The data I presented seemed to partly confirm this hypothesis, but partly also indicated that anaphors in German can be long-distance bound. To solve this paradox, I proposed that anaphors in German must always be syntactically bound, but can be assigned an interpretation by means of either semantic binding or (logophoric) co-reference. I then discussed what governs the interpretative processes through which anaphors are assigned meaning and added some tentative remarks on how a requirement on syntactic binding might follow from the hypothesis that anaphors must receive their ϕ -features from their antecedent.

Several areas could not be covered by the present study and require further research. First, I have not conclusively answered the question in which contexts semantically bound anaphors are licensed. In particular, throughout most of this study I have sidestepped some constructions for which judgments are notoriously controversial, particularly infinitives (see e.g. Gunkel 2003,

³⁰ See Preminger (2020) for a comprehensive criticism of the hypothesis that anaphor binding is underpinned by or even reduces to agreement.

Zifonun 2003) and ditransitives (see e.g. Featherstone and Sternefeld 2003). Secondly, I have ignored the intensifier *selbst* and have also largely treated all occurrences of *sich* as equals, thus ignoring the different semantic classes of verbs taking reflexive complements (see e.g. Kaufmann 2003). Taking these factors into account might allow us to refine our analysis. Thirdly, the present study, while largely eschewing syntactic movement, might have consequences for the reconstruction behaviour of German anaphors (see e.g. Georgi et al. 2019) which I have not addressed.

Finally, as for cross-linguistic comparison, I have drawn almost exclusively on English as a closely related and well-studied language that German could be confronted with. However, if the distinction between syntactic and semantic binding proves fruitful for German, it might also do so for other languages. In particular, it could help to resolve the dispute, which I touched on in chapter 2, over whether there are instances of purely syntactic long-distance binding or whether these are reducible to logophoricity. In fact, this might not be a binary choice.

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Appendix A: Materials

A.1 Experimental Items³¹

CC1 +log

- 1 Der Schauspieler präsentiert sich auf dem roten Teppich den Fotografen.
The actor presents himself on the red carpet the.DAT photographers.DAT.
The actor presents himself to the photographers on the red carpet.
- 2 Die Ministerin erwies sich als fähige Diplomatin.
The minister proved herself as capable diplomat.
The minister proved herself to be a capable diplomat.
- 3 Die Regisseurin widmet sich in ihrem neuen Film dem Konflikt im Nahen Osten.
The director devotes herself in her new film the.DAT conflict.DAT in.the Near East.
In her new film, the director devotes herself to the conflict in the Near East.
- 4 Andrej wird sich wohl nie bessern.
Andrej will himself probably never improve.
Andrej will probably never improve himself.
- 5 Laura beschäftigt sich in ihrer Forschung mit dem Begriff der Persönlichkeit.
Laura concerns herself in her research with the concept the.GEN personality.GEN.
In her research, Laura concerns herself with the concept of personality.
- 6 Die ehemalige Menschenrechtsaktivistin entwickelte sich zu einer einflussreichen Politikerin.
The former human.rights.activist developed herself to an influential politician.
The former human rights activist developed into an influential politician.
- 7 Robert erholte sich im Urlaub allmählich.
Robert recovered himself in.the vacation gradually.
Robert gradually recovered on vacation.
- 8 Lia bewegt sich wegen ihrer Verletzung nur langsam.
Lia moves herself because.of her injury only slowly.

31 Numbers of experimental items refer to the 16 token sets.

Lia moves only slowly because of her injury.

- 9 Der Machthaber stellt sich in der Öffentlichkeit als demokratischer Herrscher dar.
The ruler presents himself in the public as democratic leader PRE.
The ruler presents himself in public as a democratic leader.

- 10 Bernhard erwies sich als sehr hilfsbereit.
Bernhard proved himself as very helpful.
Bernhard proved to be very helpful.

- 11 Der Künstler widmet sich in seiner neuen Ausstellung ganz der Farbe Rot.
The artist devotes himself in his new exhibition entirely the.DAT colour.DAT red.
In his new exhibition, the artist devotes himself entirely to the colour red.

- 12 Sina hat sich sehr verändert.
Sina has herself very changed.
Sina has changed a lot.

- 13 Der Manager befasst sich mit den jüngsten Verkaufszahlen.
The manager concerns himself with the latest sales.figures.
The manager concerns himself with the latest sales figures.

- 14 Die neue Mitarbeiterin entwickelte sich bald zu einem Gewinn für das Unternehmen.
The new employee developed herself soon to a profit for the company.
The new employee soon developed into a profit for the company.

- 15 Die ehemalige Präsidentschaftskandidatin erholte sich im Urlaub allmählich
The former presidential.candidate recovered herself in.the vacation gradually
vom Wahlkampf.
from.the election.campaign.
The former presidential candidate gradually recovered from the election campaign during her vacation.

- 16 Martin entschied sich, seine Tochter zu besuchen.
Martin decided himself his daughter to visit.
Martin decided to visit his daughter.

CC1 –log

- 1 Die Statue präsentiert sich nach der Restauration dem Betrachter ganz neu.
The statue presents itself after the restoration the.DAT viewer.DAT completely new.
After the restoration the statue presents itself to the viewer in a completely new way.

- 2 Die angebliche Spionageaffäre erwies sich als Erfindung der Medien.
The alleged espionage.affair proved itself as invention the.GEN media.GEN.
The alleged espionage affair proved to be an invention of the media.
- 3 Der neue Film der Regisseurin widmet sich dem Konflikt im Nahen Osten.
The new film the.GEN director.GEN devotes itself the.DAT conflict.DAT in.the Near East.
The director's new film is dedicated to the conflict in the Near East.
- 4 Die Korruption in Italien wird sich wohl nie bessern.
The corruption in Italy will itself probably never improve.
The corruption in Italy will probably never improve.
- 5 Das Literaturfestival beschäftigt sich dieses Jahr mit dem Begriff der Persönlichkeit.
The literary.festival concerns itself this year with the concept the.GEN personality.GEN.
This year's literary festival deals with the concept of personality.
- 6 Die ehemalige Festung entwickelte sich zu einem Luxushotel.
The former fortress developed itself to a luxury.hotel.
The former fortress developed into a luxury hotel.
- 7 Die Börsenkurse erholten sich allmählich.
The stock.market.prices recovered themselves gradually.
Stock market prices gradually recovered.
- 8 Die Massendemonstration bewegt sich nur langsam.
The mass.demonstration moves itself only slowly.
The mass demonstration moves only slowly.
- 9 Die Diktatur stellt sich weiterhin als enormes Problem für den Sudan dar.
The dictatorship presents itself still as enormous problem for the Sudan PRE.
The dictatorship continues to pose an enormous problem for Sudan.
- 10 Der Raketentest erwies sich als Fehlschlag.
The missile.test proved itself as failure.
The missile test proved to be a failure.
- 11 Die neue Ausstellung des Künstlers widmet sich ganz der Farbe Rot.
The new exhibition the.GEN artist.GEN devotes itself entirely the.DAT colour.DAT red.
The artist's new exhibition is dedicated entirely to the colour red.

- 12 Die Konzerthalle hat sich durch den Umbau sehr verändert.
The concert.hall has itself through the reconstruction very changed.
The concert hall has changed a lot due to the reconstruction.
- 13 Die Studie befasst sich mit den Nebenwirkungen von Krebsmedikamenten.
The study concerns itself with the side.effects of cancer.drugs.
The study deals with the side effects of cancer drugs.
- 14 Das Projekt entwickelte sich zu politischem Zündstoff.
The project developed itself to political dynamite.
The project developed into political dynamite.
- 15 Der Regenwald erholt sich allmählich von den schweren Waldbränden.
The rainforest recovers itself gradually from the severe forest.fires.
The rainforest is gradually recovering from the severe forest fires.
- 16 Die Schlacht bei Waterloo entschied sich erst nach vielen Stunden.
The battle near Waterloo decided itself only after many hours.
The Battle of Waterloo was decided only after many hours.

CC2 +log

- 1 Das Bild von dem Schauspieler gefällt sich gut.
The picture of the actor pleases himself well.
The picture of the actor pleases himself.
- 2 Die fragwürdige Geschichte über die Ministerin ärgerte sich.
The dubious story about the minister bothered herself.
- 3 Der im Feuilleton erschienene Essay über die Regisseurin freute sich.
The in.the feuilleton published essay about the director pleased herself.
The essay about the director that was published in the feuilleton pleased herself.
- 4 Der Ermittlungsbericht über Andrej beunruhigt sich.
The investigation.report about Andrej worries himself.
The investigation report about Andrej worries himself.
- 5 Die Anschuldigungen gegen Laura störten sich.
The accusations against Laura bothered herself.
- 6 Der versuchte Anschlag auf die Menschenrechtsaktivistin belastete sich emotional.
The attempted attack on the human.rights.activist burdened herself emotionally.
The attempted attack on the human rights activist was an emotional burden on herself.

- 7 Die Beurteilung von Robert erschien sich fair.
The evaluation of Robert seemed himself fair.
The evaluation of Robert seemed fair to himself.
- 8 Die Doku über Lia amüsiert sich.
The documentary about Lia amuses herself.
- 9 Der entschlossene Kampf gegen den Machthaber alarmiert sich.
The determined fight against the ruler alarms himself.
- 10 Der gehässige Witz über Bernhard belustigte sich nur.
The spiteful joke about Bernhard amuses himself only.
The spiteful joke about Bernhard only amuses himself.
- 11 Das Lob für den Künstler freute sich.
The praise for the artist pleased himself.
- 12 Das kritische Gutachten über den Vorsitzenden macht sich große Sorgen.
The critical report on the chairman makes himself.DAT great concerns.
The critical report on the chairman causes himself great concern.
- 13 Die Ermittlungen gegen den Manager störten sich.
The investigations against the manager bothered himself.
- 14 Der Einsatz für die neue Mitarbeiterin rührte sich.
The enthusiasm for the new employee touched herself.
- 15 Die Petition gegen die Präsidentschaftskandidatin entmutigte sich.
The petition against the presidential.candidate discouraged herself.
The petition against the presidential candidate discouraged herself.
- 16 Die Doku über Martin amüsiert sich.
The documentary about Martin amuses himself.

*CC2 -log*³²

- 1 Das Bild von der Statue zeigt sich nach der Restauration.

32 As discussed in chapter 3.4, the CC2 +log sentences allow for two readings, both of which are ungrammatical:

- (i) The accusations in the media against [the literary festival]_i interrupted itself_i.
- (ii) [The accusations in the media against the literary festival]_i interrupted themselves_i.

Whenever the ambiguity is reflected in glosses and translations, I will only give the intended reading in (i).

- The picture of the statue shows itself after the restoration.
The picture of the statue shows itself after restoration.
- 2 Die fragwürdige Geschichte über die Spionageaffäre brachte sich in alle Medien.
The dubious story about the espionage.affair got itself into all media.
The dubious story about the espionage affair got itself into the media.
- 3 Der im Feuilleton erschienene Essay über den neuen Film verreit sich.
The in.the feuilleton published essay about the new film tore.up itself.
The essay about the new film that was published in the feuilleton heavily criticised itself.
- 4 Der Ermittlungsbericht über die Korruption in Italien dokumentiert sich bis ins Detail.
The investigation.report about the corruption in Italy documents itself to in.the detail.
The investigation.report about the corruption in Italy documents itself in every detail.
- 5 Die Anschuldigungen gegen das Literaturfestival in den Medien unterbrachen sich.
The accusations against the literary.festival in the media interrupted itself.
The accusations in the media against the literary festival interrupted itself.
- 6 Die Anschlag auf die Festung zerstörte sich.
The attack on the fortress destroyed itself.
- 7 Die Beurteilung von den Börsenkursen stellte sich im besten Licht dar.
The evaluation of the stock.market.prices presented themselves in.the best light PRE.
The evaluation of the stock market prices presented themselves in the best light.
- 8 Die Doku über die Massendemonstration fasst sich in wenigen Minuten zusammen.
The documentary about the mass.demonstration summarises itself in a.few minutes
PRE.
The documentary about the mass demonstration summarises itself in a few minutes.
- 9 Der entschlossene Kampf gegen die Diktatur im Sudan schwächt sich.
The determined fight against the dictatorship in.the Sudan weakens itself.
The determined fight against the dictatorship in Sudan weakens itself.
- 10 Der Witz über den fehlgeschlagenen Raketentest verspottete sich.
The joke about the failed missile.test mocked itself.
The joke about the failed missile test mocked itself.
- 11 Das Lob für die Ausstellung wurde sich völlig gerecht.
The praise for the exhibition became itself.DAT fully just.
The praise for the exhibition did itself full justice.
- 12 Das kritische Gutachten über die geplante neue Konzerthalle wirft sich vor allem

The critical report on the planned new concert hall accuses itself above all of high costs.

high costs PRE.

The critical report on the planned new concert hall accuses itself above all of high costs.

- 13 Die Ermittlungen gegen den Pharmakonzern ruinierten sich.

The investigations against the pharmaceutical company ruined itself.

The investigations against the pharmaceutical company ruined itself.

- 14 Der mangelnde Einsatz für das Projekt verurteilte sich zum Scheitern.

The lacking enthusiasm for the project condemned itself to the failure.

The lack of enthusiasm for the project condemned itself to failure.

- 15 Die Petition für den Regenwald soll sich vor illegaler Rodung schützen.

The petition for the rainforest aims to protect itself from illegal clearing.

The petition for the rainforest aims to protect itself from illegal clearing.

- 16 Die Doku über die Schlacht bei Waterloo fasst sich in wenigen Minuten

The documentary about the battle near Waterloo summarises itself in a few minutes together.

PRE.

The documentary about the Battle of Waterloo summarises itself in a few minutes.

TC1 +log

- 1 Das Foto von sich gefällt dem Schauspieler gut.

The photo of himself pleases the actor well.

The photo of himself pleases the actor.

- 2 Das haltlose Gerücht über sich ärgerte die Ministerin.

The baseless rumour about herself bothered the minister.

- 3 Der beifällige Artikel über sich freute die Regisseurin.

The positive article about herself pleased the director.

- 4 Die investigative Reportage über sich beunruhigt Andrej.

The investigative reportage about himself worries Andrej.

- 5 Die zunehmende Kritik an sich störte Laura.

The increasing criticism of herself bothered Laura.

- 6 Die Angriffe auf sich belasteten die Menschenrechtsaktivistin emotional.

The attacks on herself burdened the human rights activist emotionally.

The attacks on herself were an emotional burden on the human rights activist.

- 7 Die Bewertung von sich erschien Robert fair.

The evaluation of himself seemed Robert.DAT fair.

The evaluation of himself seemed fair to Robert.

- 8 Das Video über sich amüsiert Lia.

The video about herself amused Lia.

- 9 Der wachsende Widerstand gegen sich alarmiert den Machthaber.

The growing resistance against himself alarms the ruler.

- 10 Der spöttische Kommentar über sich belustigte Bernhard nur.

The mocking comment about himself amused Bernhard only.

The mocking comment about himself only amused Bernhard.

- 11 Die positive Rezension über sich freute den Künstler.

The positive review about himself pleased the artist.

- 12 Das kritische Gutachten über sich macht dem Vorsitzenden große Sorgen.

The critical report on himself makes the.DAT chairman.DAT great concerns.

The critical report on himself causes the chairman great concern.

- 13 Die Vorwürfe gegen sich störten den Manager.

The accusations against himself bothered the manager.

- 14 Die ablehnende Haltung sich gegenüber bedrückte die neue Mitarbeiterin.

The hostile attitude herself towards depressed the new employee.

The new employee was depressed by the hostile attitude towards herself.

- 15 Die Pressekampagne gegen sich entmutigte die Präsidentschaftskandidatin.

The press.campaign against herself discouraged the presidential.candidate.

The press campaign against herself discouraged the presidential candidate.

- 16 Die Erzählung über sich amüsiert Martin.

The story about himself amused Martin.

TC1 –log

- 1 Das Foto von sich zeigt die Statue nach der Restauration.

The photo of itself shows the statue after the restoration.

The photo of itself shows the statue after restoration.

- 2 Das haltlose Gerücht über sich brachte die Spionageaffäre in alle Medien.

The baseless rumour about itself got the spying.affair into all media.

- The baseless rumour about itself got the spying affair into all the media.
- 3 Der vernichtende Artikel über sich verriss den Film.
The scathing article about itself tore apart the film.
The scathing article about itself heavily criticised the film.
- 4 Die investigative Reportage über sich dokumentiert die Korruption in Italien bis ins Detail.
The investigative reportage about itself documents the corruption in Italy to in.the detail.
The investigative reportage about itself documents the corruption in Italy in every detail.
- 5 Die zunehmende Kritik an sich in den Medien unterbrach das Literaturfestival.
The increasing criticism of itself in the media interrupted the literature festival.
The increasing criticism of itself in the media interrupted the literature festival.
- 6 Die Angriffe auf sich zerstörten die Festung allmählich.
The attacks on itself destroyed the fortress gradually.
The attacks on itself gradually destroyed the fortress.
- 7 Die Bewertung von sich stellte die Börsenkurse im besten Licht dar.
The evaluation of themselves presented the stock market prices in the best light PRE.
The evaluation of themselves presented the stock market prices in the best light.
- 8 Das Video über sich fasst die Massendemonstration in wenigen Minuten zusammen.
The video about itself summarises the mass demonstration in a few minutes PRE.
The video about itself summarises the mass demonstration in a few minutes.
- 9 Der wachsende Widerstand gegen sich schwächt die Diktatur im Sudan.
The growing resistance against itself weakens the dictatorship in the Sudan.
The growing resistance against itself weakens the dictatorship in Sudan.
- 10 Der abschätzige Kommentar über sich verspottete den fehlgeschlagenen Raketentest.
The derogatory comment about itself mocked the failed missile test.
The derogatory comment about itself mocked the failed missile test.
- 11 Die positive Rezension über sich wurde der Ausstellung völlig gerecht.
The positive review about itself became the DAT exhibition. DAT fully just.
The positive review about itself did the exhibition full justice.
- 12 Das kritische Gutachten über sich wirft der geplanten neuen Konzerthalle vor allem hohe Kosten vor.
The critical report on itself accuses of the DAT planned. DAT new. DAT concert hall. DAT above all high costs PRE.

The critical report on itself accuses the planned new concert hall above all of high costs.

- 13 Die Vorwürfe gegen sich ruinierten den Pharmakonzern.

The accusations against itself ruined the pharmaceutical company.

The accusations against itself ruined the pharmaceutical company.

- 14 Die ablehnende Haltung sich gegenüber verurteilte das Projekt zum Scheitern.

The hostile attitude itself towards condemned the project to the failure.

The hostile attitude towards itself condemned the project to failure.

- 15 Die Kampagne für sich soll den Regenwald vor illegaler Rodung schützen.

The campaign for itself aims to protect the rainforest from illegal clearing.

The campaign for itself aims to protect the rainforest from illegal clearing.

- 16 Die Erzählung über sich fasst die Schlacht bei Waterloo auf wenigen Seiten

The story about itself summarises the battle near Waterloo on a few pages
zusammen.

PRE.

The story about itself summarises the Battle of Waterloo on a few pages.

TC2 +log

- 1 Der Schauspieler ist überzeugt, dass das angebliche Foto von sich eine Fälschung ist.

The actor is convinced that the alleged photo of himself is a fake.

The actor is convinced that the alleged photo of himself is a fake.

- 2 Die Ministerin stellte mit Erschrecken fest, dass die Gerüchte über sich sogar in seriösen

The minister noted with dismay PRE that the rumors about herself even in serious
Zeitungen diskutiert wurden.

newspapers discussed were.

The minister noted with dismay that the rumors about herself were even discussed in serious
newspapers.

- 3 Die Regisseurin war bestürzt, dass der Artikel über sich kein gutes Haar an ihr gelassen

The director was shocked that the article about herself no good hair on her left
hatte.

had.

The director was shocked that the article about herself had nothing good to say about her.

- 4 Andrej hofft, dass die investigative Reportage über sich nie veröffentlicht wird.

Andrej hopes that the investigative reportage about himself never published is.

Andrej hopes that the investigative reportage about himself will never be published.

- 5 Laura litt darunter, dass die Kritik an sich immer lauter wurde und ihr Laura suffered from this that the criticism of herself increasingly louder became and her Selbstwertgefühl untergrub.

self-esteem undermined.

Laura suffered from the fact that the criticism of herself became louder and louder and undermined her self-esteem.

- 6 Die Menschenrechtsaktivistin hatte Angst, dass die Angriffe auf sich zunehmen würden. The human rights activist had fear that the attacks on herself increase would. The human rights activist was afraid that the attacks on herself would increase.

- 7 Robert war froh, dass die Bewertung von sich so positiv ausgefallen war. Robert was glad that the evaluation of himself so positive turned out was. Robert was glad that the evaluation of himself had turned out so positive.

- 8 Lia war erstaunt, dass das Video von sich von Tausenden gesehen worden war. Lia was surprised that the video of herself by thousands seen been was. Lia was surprised that the video of herself had been seen by thousands.

- 9 Der Machthaber ist überzeugt, dass der Widerstand gegen sich nur wenig Erfolg haben will. The ruler is convinced that the resistance against himself only little success have will. wird.

The ruler is convinced that the resistance against himself will have little success.

- 10 Bernhard stellte mit Unbehagen fest, dass der abschätzigste Kommentar über sich Bernhard noted with uneasiness that the derogatory comment about himself überall verbreitet wurde. everywhere spread was.

Bernhard noticed with uneasiness that the derogatory comment about himself was spread everywhere.

- 11 Der Künstler war bestürzt, dass die Rezension über sich so negativ ausgefallen war. The artist was dismayed that the review about himself so negative turned out was. The artist was dismayed that the review about himself had turned out so negative.

- 12 Der neue Vorsitzende hofft, dass die anfänglichen Zweifel an sich bald ausgeräumt sein werden. The new chairman hopes that the initial doubts about himself soon dispelled be will.

The new chairman hopes that the initial doubts about himself will soon be dispelled.

- 13 Der Manager litt darunter, dass die Vorwürfe gegen sich immer lauter
The manager suffered from this that the accusations against himself increasingly louder
wurden und seine Arbeit nicht wertgeschätzt wurde.
became and his work not appreciated was.

The manager suffered from the fact that the accusations against himself became louder and louder and that his work was not appreciated.

- 14 Die neue Mitarbeiterin hatte Angst, dass die Haltung sich gegenüber eher ablehnend sein
The new employee had fear that the attitude herself towards rather hostile be
würde.
would.

The new employee was afraid that the attitude towards herself would be rather hostile.

- 15 Die Präsidentschaftskandidatin war erleichtert, dass die Pressekampagne gegen sich nur
The presidential candidate was relieved that the press campaign against herself only
wenig Unterstützer fand.
few supporters found.

The presidential candidate was relieved that the press campaign against herself found only few supporters.

- 16 Martin war erstaunt, dass die Erzählung über sich so bekannt war.
Martin was surprised that the story about himself so well-known was.
Martin was surprised that the story about himself was so well-known.

TC2 –log

- 1 Die lang verschollene, nun wiedergefundene Statue beweist, dass das angebliche Foto von
The long lost, now rediscovered statue proves that the alleged photo of
sich eine Fälschung sein muss.
itself a fake be must.

The long-lost, now rediscovered statue proves that the alleged photo of itself must be a fake.

- 2 Die obskure Spionageaffäre hatte zur Folge, dass die Gerüchte über sich sogar in
The obscure espionage affair had to the consequence that the rumours about itself even in
seriösen Zeitungen diskutiert wurden.
serious newspapers discussed were.

The obscure espionage affair had the consequence that the rumors about itself were even discussed in serious newspapers.

- 3 Der Film verdient es wirklich, dass der Artikel über sich kein gutes Haar an ihm gelassen hat.
The film deserves it really that the article about itself no good hair on itself left hat.
has.

The film really deserves it that the article about itself had nothing good to say about it.

- 4 Die Korruption wird verhindern, dass die investigative Reportage über sich tatsächlich veröffentlicht wird.
The corruption will prevent that the investigative reportage about itself actually published is.

Corruption will prevent the investigative reportage about itself from actually being published.

- 5 Das Literaturfestival litt darunter, dass die Kritik an sich immer lauter wurde und die Besucher ausblieben.
The literature.festival suffered from.this that the criticism of itself increasingly louder became and the visitors stayed.away.

The literature festival suffered from the fact that the criticism of itself became louder and louder and the visitors stayed away.

- 6 Die ehemalige Festung erzählt noch heute davon, dass die Angriffe auf sich am Ende des Mittelalters zunahmen.
The former fortress tells still today of.this that the attacks on itself at.the end the.GEN Middle.Ages.GEN increased.

The former fortress still tells us that the attacks on itself increased at the end of the Middle Ages.

- 7 Die Börsenkurse profitierten davon, dass die Bewertung von sich so positiv ausgefallen war
The stock.market.prices profited from.this that the evaluation of themselves so positive turned.out was.

Stock market prices profited from the fact that the evaluation of themselves had turned out so positive.

- 8 Die Massendemonstration bewirkte, dass das Video von sich von Tausenden gesehen wurde.
The mass.demonstration had.the.effect that the video of itself by thousands seen was.

The mass demonstration had the effect that the video of itself was seen by thousands.

- 9 Die weiterhin bestehende Diktatur im Sudan beweist, dass der Widerstand gegen sich
The still existing dictatorship in.the Sudan proves that the resistance against itself
nur wenig Erfolg hatte.
only little success had.

The continuing dictatorship in Sudan proves that the resistance against itself had little success.

- 10 Der fehlgeschlagene Raketentest hatte zur Folge, dass der abschätzigste Kommentar
The failed missile.test had to.the consequence that the derogatory comment
über sich überall verbreitet wurde.
about itself everywhere spread was.

The failed missile test had the consequence that the derogatory comment about itself was spread everywhere.

- 11 Die Ausstellung verdient es wirklich, dass die Rezension über sich so negativ ausgefallen
The exhibition deserves it really that the review about itself so negative turned.out
ist.
is.

The exhibition really deserves it that the review about itself turned out so negative.

- 12 Die gelungene neue Konzerthalle beweist, dass die anfänglichen Zweifel an sich
The successful new concert.hall proves that the initial doubts about itself
ungerechtfertigt waren.
unjustified were.

The successful new concert hall proves that the initial doubts about itself were unjustified.

- 13 Der Pharmakonzern litt darunter, dass die Vorwürfe gegen sich
The pharmaceutical.company suffered from.this that the accusations against itself
immer lauter wurden und die Aufträge ausblieben.

Increasingly louder became and the orders failed.to.come.in.

The pharmaceutical company suffered from the fact that the accusations against itself became louder and louder and orders failed to come in.

- 14 Das erfolglose Projekt zeigt, dass die Haltung sich gegenüber eher ablehnend war.
The unsuccessful project shows that the attitude itself towards rather hostile was.
The unsuccessful project shows that the attitude towards itself was rather hostile.

- 15 Der Regenwald profitierte davon, dass die Kampagne für sich zahlreiche Unterstützer
The rainforest profited from.this that the campaign for itself numerous supporters
fand.

found.

The rainforest profited from the fact that the campaign for itself found numerous supporters.

- 16 Die Schlacht bei Waterloo bewirkte, dass die Erzählung über sich noch heute bekannt
The battle near Waterloo had the effect that the story about itself still today known
ist.

is.

The Battle of Waterloo had the effect that the story about itself is still known today.

A.2 Filler Items

Grammatical filler items

- 101 Michael ging gerne zur Schule.
Michael went with pleasure to the school.
Michael liked going to school.
- 102 Maisha erschrak, als sie ihren Bruder sah.
Maisha was shocked when she her brother saw.
Maisha was shocked when she saw her brother.
- 103 Die Großmutter vergaß nie die Geburtstage ihrer Enkel.
The grandmother forgot never the birthdays her GEN grandchildren GEN.
The grandmother never forgot her grandchildren's birthdays.
- 104 Das Ministerium veröffentlichte eine Erklärung.
The Ministry published a statement.
- 105 Seit vielen Monaten war der See zugefroren.
For many months was the lake frozen over.
For many months the lake was frozen over.
- 106 Rachel würde gerne im Urlaub ans Meer fahren.
Rachel would with pleasure in the vacation to the sea go.
Rachel would like to go to the sea during her vacation.
- 107 In diesem Gebäude ist Rauchen verboten.
In this building is smoking prohibited.
Smoking is prohibited in this building.
- 108 Die Zeitung hat einen Bericht über Kernkraftwerke veröffentlicht.
The newspaper has an article on nuclear power plants published.

The newspaper has published an article on nuclear power plants.

109 Die Angestellten werden ab morgen wieder zur Arbeit kommen.

The employees will from tomorrow again to work come.

From tomorrow the employees will return to work.

110 Die Trainerin sagte das Spiel ab, da das Unwetter immer stärker wurde.

The coach cancelled the game PRE because the storm increasingly stronger became.

The coach cancelled the game because the storm was getting stronger and stronger.

111 Der Student bemüht sich, die Professorin zu verstehen.

The student tries himself the professor to understand.

The student tries to understand the professor.

112 Obwohl sie müde war, ging Alina noch nicht ins Bett.

Although she tired was went Alina yet not in.the bed.

Although she was tired, Alina did not go to bed yet.

113 Niemand wusste, dass es bereits angefangen hatte zu schneien.

Nobody knew that it already started had to snow.

Nobody knew that it had already started snowing.

114 Der Wald stand in Flammen.

The forest stood in flames.

The forest was in flames.

115 Das Haus war vor kurzem renoviert worden.

The house was before shortly renovated been.

The house had recently been renovated.

116 Die Straße führt erst an der Moschee vorbei und dann aus dem Dorf hinaus.

The road leads first at the mosque past and then out.of the village out.

The road leads first past the mosque and then out of the village.

Intermediate filler items

117 Das Becken des Brunnens kann man mit seinem eigenen Wasser

The.ACC basin.ACC the.GEN fountain.GEN can one with its own water
reinigen.

clean.

One can clean the basin of the fountain with its own water.

118 Der Erzieher notiert, wer heute fehlt von den Kindern.

The teacher notes who today is missing of the children.

The teacher notes who of the children is missing today.

119 Den Gewinner mit einer Urkunde ausgezeichnet hat man.

The ACC winner ACC with a certificate awarded has one.

One has awarded the winner with a certificate.

120 Die Wissenschaftlerin hat das Argument widerlegt, dass Zucker gesund sei, das ihr

The scientist has the argument refuted that sugar healthy is which her
Kollege angeführt hatte.

colleague put forward had.

The scientist refuted the argument that sugar is healthy, which her colleague had put
forward.

121 Rahul erhielt immer gute Noten, ohne viel gelernt dafür zu haben.

Rahul got always good grades without much learned for it to have.

Rahul always got good grades without having learned much for it.

122 Dieser Berg zieht viele Wanderer an da sein eigener Gipfel die der

This mountain attracts many hikers PRE because its own peak those the GEN
anderen Berge überragt.

other GEN mountains GEN overlooks.

This mountain attracts many hikers because its own peak overlooks those of the other
mountains.

123 Beate war erleichtert, dass endlich jemand ihr sie vorstellte.

Beate was relieved that finally someone her DAT her ACC introduced.

Beate was relieved that someone finally introduced her to her.

124 Sie hatte gedacht, dass heutzutage dass Männer und Frauen gleichberechtigt sind niemand

She had thought that nowadays that men and women equal are nobody
seriously doubted.

ernsthaft bezweifelte.

She had thought that nowadays nobody seriously doubted that men and women are equal.

125 Eine Postkarte wollte Ali schreiben an seine Mutter.

A ACC postcard ACC wanted Ali write to his mother.

Ali wanted to write a postcard to his mother.

126 Ihr Freund fragte sie, wen es sie gefreut hätte, auf der Party zu treffen.

Her friend asked her who ACC it her ACC pleased would have at the party to meet.

Her friend asked her who it would have pleased her to meet at the party.

- 127 Das Freilassen im Wald von Hunden ist untersagt.
 The release in.the forest of dogs is prohibited.
 It is prohibited to release dogs in the forest.
- 128 Sie erinnerte sich nicht mehr, wann was wer beobachtet hatte.
 She remembered herself not anymore when what who observed had.
 She no longer remembered when who had observed what.
- 129 Sorgfältig hat Jean glücklicherweise den Rasen gemäht.
 Carefully has Jean fortunately the lawn mowed.
 Fortunately, Jean mowed the lawn carefully.
- 130 Die Regierung entscheidet heute, welche Meinung sie vertreten wird über den
 The government decides today what opinion it hold will on the
 Handelskrieg.
 trade.war.
 Today the government decides what opinion it will hold on the trade war.
- 131 Marlene versuchte, etwas jemandem zu erklären.
 Marlene tried something someone.DAT to explain.
 Marlene tried to explain something to someone.
- 132 Jeder ärgert sich über sich selbst manchmal.
 Everyone gets.annoyed themselves about themselves self sometimes.
 Everyone gets annoyed with themselves sometimes.

Ungrammatical filler items

- 133 Mein Bruder und seine Frau ist wieder einmal zu spät.
 My brother and his wife is again once too late.
- 134 Der Vater fragt sich, wie sein Sohn nicht gerne Spaghetti isst.
 The father asks himself how his son not with.pleasure Spaghetti eats.
- 135 Der Personalmanager hat sie gefunden für ihren Job ungeeignet.
 The HR.manager has her found for her job unsuitable.
- 136 Der Hund ist schlafend.
 The dog is sleeping.
- 137 Der Weg durch dem Wald ist am kürzesten.
 The path through the.DAT forest.ACC is the shortest.
- 138 Betty war nicht bewusst, über was ihr Kollege gekündigt hatte wegen einem Streit.

- Betty was not aware about what her coworker quit had because of a fight.
- 139 Er hat den Apfel geschnitten in kleine Stücke.
He has the apple cut into small pieces.
- 140 Die Schülerin ging schnell zur nächsten Haltestelle.
The pupil went quickly to the next stop.
- 141 Jeder der Anwesenden wussten, wer gemeint war.
Everyone the present knew who meant was.
- 142 Die Lösung würde niemand glauben, dass er gefunden hatte.
The solution would nobody believe that he found had.
- 143 Schnell gerne fährt Alessandra Auto.
Fast with pleasure drives Alessandra car.
- 144 Andreas überlegte, wer sein Freund gesagt hatte, dass die Geschichte erzählt hat.
Andreas wondered who his friend said had that the story told had.
- 145 Da steht ein paar Bücher im Regal.
There stands a few books in the bookshelf.
- 146 Erik beschloss, dass er nebeneinander die Stühle stellen würde.
Erik decided that he next to each other the chairs put would.
- 147 Tanya wusste nicht, wessen ihre Freundin Gitarre gespielt hatte.
Tanya knew not whose her friend guitar played had.
- 148 Niemand kann arbeiten immer perfekt.
Nobody can work always perfectly.

Appendix B: Results

B.1 Experimental Results

Rating	CC1 +log	CC1 -log	CC2 +log	CC2 -log	TC1 +log	TC1 -log	TC2 +log	TC2 -log
1	3	8	216	159	89	156	118	144
2	4	4	39	54	48	57	66	71
3	7	11	5	22	41	30	30	27
4	30	34	3	19	47	12	27	18
5	218	207	0	10	39	9	23	4

Table 3: Distribution of ratings per variable level combination

B.2 Demographic Data

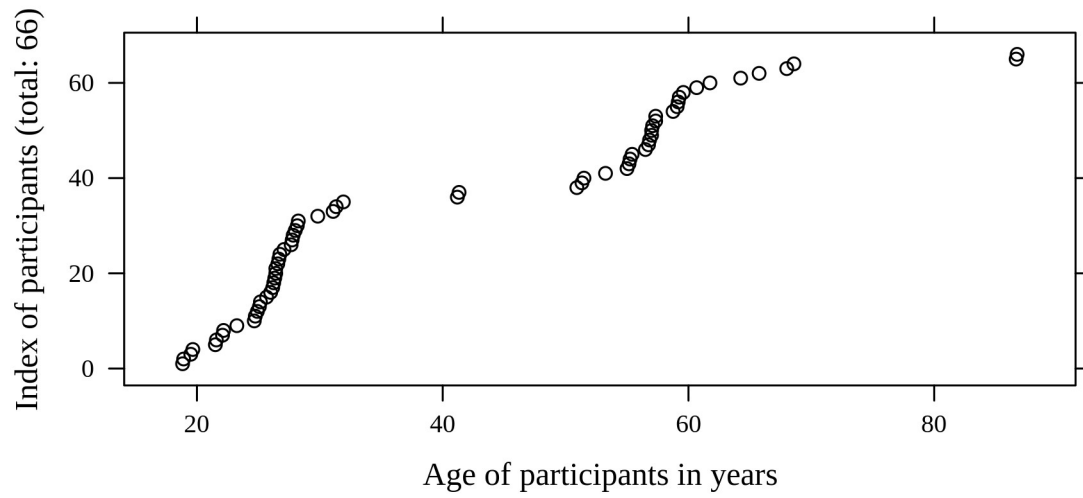


Figure 2: Age distribution of participants

Place of origin	Absolute numbers	Percentage
North Rhine-Westphalia	42	≈ 64%
Baden-Wuerttemberg	4	≈ 6%
Saarland	4	≈ 6%
Berlin	3	≈ 5%
Hesse	3	≈ 5%
Lower Saxony	3	≈ 5%
Rhineland-Palatinate	3	≈ 5%
Schleswig-Holstein	2	≈ 3%
Saxony	1	≈ 2%
other	1	≈ 2%

Table 4: Geographic origin of participants